F24-175-D-PulseViz: ECG ReImagined

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Chapter 1

Introduction

The purpose of this document is to provide an insight about the implementation and docu- mentation of PulseViz, an AI powered mobile application to analyze ECG (Electrocardio- gram) scans and detect heart attack in real-time. This document provides a comprehensive overview of the whole system, including the objectives, features, problems it addresses, stakeholders and technical details like timelines and the technologies required to provide a clear understanding of the project.

The current healthcare system in Pakistan is not up to the minute, we lack tools and technologies for timely detection of cardiac emergencies, which is a leading cause of fatalities. The lack of advanced diagnostic tools , limited availability of healthcare experts and extensive ECG reading process make the analysis process difficult and increases the risk of lives.

PulseViz - a system that provides solutions to all of these problems by leveraging AI to provide real-time and automated analysis of the ECG scans. The mobile app is developed to analyze ECG and detect heart attacks, sending instant alerts to healthcare profession- als to ensure timely treatment. Saving lives with each scan.

1.1 Problem Statement

Patients in Pakistan face a lot of issues regarding health care as there are delays in the diagnosis of diseases and treatment. This results in detrimental conditions for patients and in the worst case, even death. The main problems encountered in hospitals are fewer medical experts and the lengthy process of interpreting the ECG (Electrocardiogram) manually. It is a time-consuming task which further aggravates the patient's health. Our app aims to provide a solution to this problem through an AI-powered, real-time ECG (Electrocar- diogram) analysis. Automating ECG (Electrocardiogram) readings will make the whole system less dependent on specialized medical personnel and accelerate the whole process.

It will make the detection of heart attacks more smooth and accurate. Personalized health reports will be generated and can be accessed remotely through cloud integration to assist healthcare providers. Through this system, hospitals can respond faster in critical situations. It relieves the healthcare staff of workload. Finally, our solution will enhance the whole process of heart attack detection in resource-constrained hospitals, saving lives and streamlining patient management.

1.2 Scope

The proposed project is an AI-powered solution designed to detect heart attacks in real time by analyzing scanned Electrocardiogram (ECG) images. This project is being devel- oped specifically for hospitals in Pakistan, ensuring compatibility with existing hospital infrastructure. The goal is to allow professionals to access and utilize the tool without disrupting current workflows.

The system will employ advanced machine learning algorithms to detect patterns and abnormalities in ECG scans that could indicate potential cardiac problems. By analyzing ECG data in real-time, the app will provide immediate feedback on heart attack risks. A real-time alert system will notify relevant healthcare teams in case of an emergency. Additionally, the app will generate a report for each patient at the end of the session, helping healthcare professionals in making informed decisions and treat the infected in time.

The app will feature a simple user-friendly interface, designed for ease of use by medical staff such as doctors and nurses. All patient data entered into the app will be securely stored in a database, in abidance with local and international healthcare data regulations. The primary aim of this app is to reduce diagnostic delays and enhance the timely and effective detection of heart attacks.

1.3 Modules

Our application is a mobile application with the following modules:

1.3.1 Module 1: Electrocardiogram (ECG) Scanning

In this module, the user can scan the ECG images through their mobile phone's camera, our app will process them, and get an analysis of the heart activity.

1. ECG images are scanned and uploaded through the mobile app.

2. Multiple images are taken of the ECG paper depending upon the recording time of the heart of the patient.

1.3.2 Module 2: Heart Attack Detection

In this module the app will detect the occurrence of heart attack by analyzing the ECG images, through Artificial Intelligence, providing real time feedback.

1. Analysis of scanned images in real-time thus providing instant feedback.

2. Detection of heart abnormalities, including heart attack.

1.3.3 Module 3: Personalized Health Reports

This app module generates personalized reports of patients, allowing them to get an idea of their heart health.

1. Generate personalized health reports and keep track of heart health.

2. A comprehensive and brief report with crucial information depicting cardiac health.

1.3.4 Module 4: Generate Notifications and Alerts

In this module the app generates and sends real-time notifications and alerts in case of emergencies to the concerned respective.

1. Send real-time notifications and alerts in case of emergencies.

2. The doctor/heart specialist is notified in case of emergency.

1.3.5 Module 5: View Patient History

In this module, the user, i.e., doctors and nurses, can view a patient’s history and previous records in case the patient visits again.

1. View patient’s previous records and reports.

2. Compare the patient's progress and history for present performance evaluation.

1.3.6 Module 6: Cloud Integration and Data Management

This module enhances and manages the storing and retrieving of ECG and personal data. The cloud is integrated to ensure the accessibility of data from anywhere while maintain- ing safety in accordance with data privacy standards.

1. Management and storing of user ECG data in the cloud.

2. Access control for sensitive data.

1.4 User Classes and Characteristics

Our user classes that will use this product are nurse and doctor. Their pertinent characteristics are stated below:

User Class	Description
Nurse	The nurse is one of the main users of the system. All the nurses of the hospitals will be using this system. The nurse should be experienced in patient care and possess basic technical knowledge. The nurse will use PulseViz to scan images, view results and reports, and take further actions based on the results. The nurse requires PulseViz to be extremely user-friendly and easy to use. Additionally, the nurse will use PulseViz to save reports against specific patients in the system's database for later use.
Doctor	The doctor is another main user of the system. All the doctors at the hospital will fall into this category. Doctors must be experienced in healthcare and have basic technical proficiency. They are the main entities treating patients. Doctors will use PulseViz to scan images, view results and reports, and take further actions based on the results. In case of an emergency, available doc- tors will be notified with results and reports attached. Doctors will also use PulseViz to save patient reports for later use. They require the app to be fast, efficient, and user-friendly.

Chapter 2

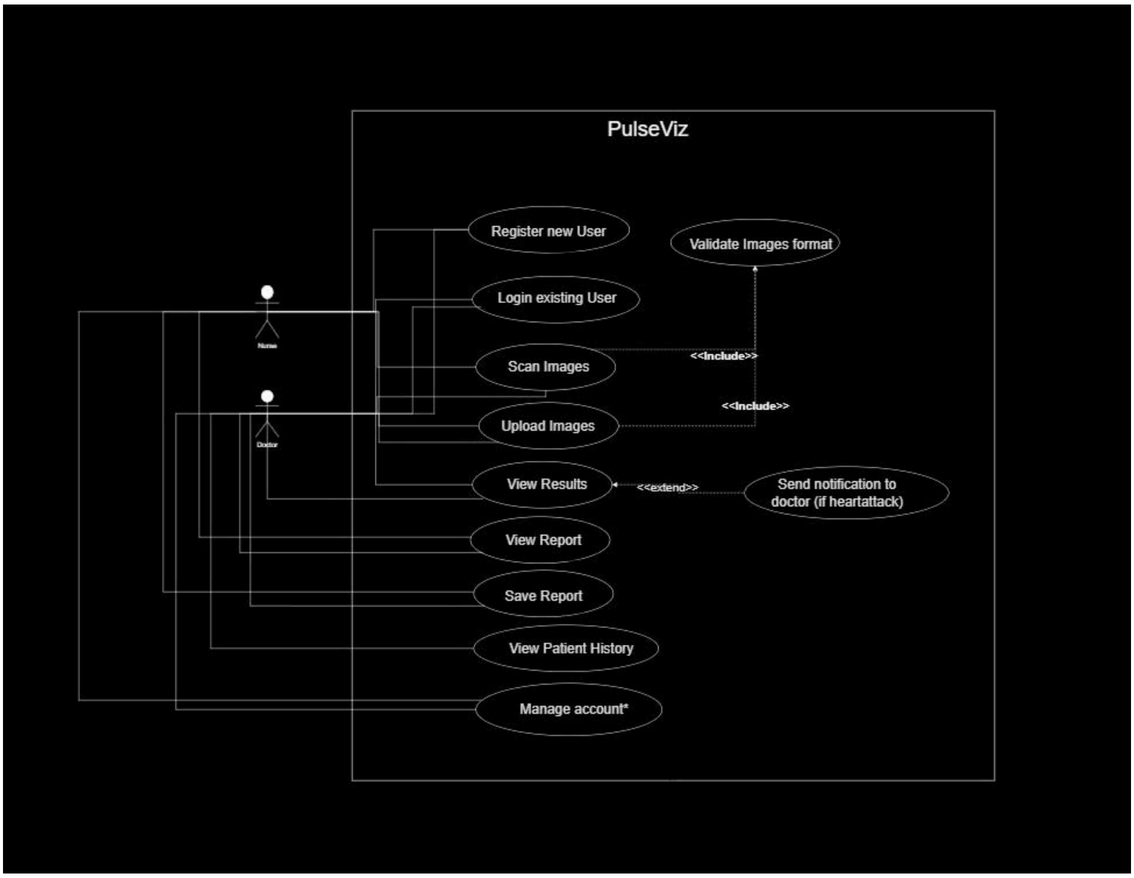
Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

The Use Case diagram depicting our app's end user's interaction is provided below:

2.1.1 Use Case Diagram



2.1.2 Extended Use Cases

2.1.2.1 Use Case Name : Register New User

Primary Actor: Nurse,Doctor

Scope: PulseViz App

Level: Primary

Stakeholders & Interests:

Nurse : Nurse wants to be able to register onto the app without any hurdles , so they can start using the app efficiently .

Doctor : Doctor wants to be able to register onto the app without any hurdles successfully, so they can start using the app efficiently.

Hospital : The hospital wants that all of their doctors and nurses can successfully register on the app so the app can actually be useful to their existing system.

Preconditions: None

Postconditions: The user (nurse or the doctor) is successfully registered on the app and their respective data has been stored in the database successfully.

Main Success Scenario:

User Action	System Response
1 - User opens the mobile app	
2- User is redirected to the register screen automatically.	
3. Users enter their credentials and hit Register.	
	4. The system verifies the credentials.
	5. The system successfully registers the user and stores data in the database.
	6.System shows a successful snackbar to the user.
7.The user is now redirected to the main screen and is ready to use the app.	

Extensions :

1. At any time the system crashes : The user is then obliged to restart the app and go through the registration process again.

2. If the user already exists : If a user with the given credentials already exists then

the user must be told that their id already exists and they should login instead of registering all over again.

2.1.2.2 Use case name : Login existing user

Primary Actor: Nurse,Doctor

Scope: PulseViz App

Level: Primary

Stakeholders & Interests:

Nurse : Nurse wants to be able to login onto the app without any hurdles , so

they can start using the app efficiently .

Doctor : Doctor wants to be able to login onto the app without any hurdles successfully, so they can start using the app efficiently.

Hospital : The hospital wants that all of their doctors and nurses can successfully login on the app so the app can actually be useful to their existing system.

Preconditions: The user has successfully been registered.

PostConditions: The user (nurse or the doctor) is successfully logged in on the app and they can now use the app for detection purposes.

Main Success Scenario :

User Action	System Response
1 - User opens the mobile app	
2- User is redirected to the login screen automatically.	
3. User enters their credentials and Login.	
	4. The system verifies the credentials.
	5. The system successfully logs in the user and gives them access to the app.
	6.System shows success snackbar to the user and redirect them to the home screen.
7.The user is now ready to use the app.	

Extensions :

1. At any time the system crashes : The user is then obliged to restart the app and go through the registration process again.
2. Incorrect Credentials : If a user enters incorrect credentials , the login process fails and the user is asked to input the correct credentials.

2.1.2.3 Use case name : Scan Images

Primary Actor: Nurse,Doctor

Scope: PulseViz App

Level: Primary

Stakeholders & Interests:

Nurse : Nurse wants to be able to scan pictures of the ECG's and pass them to the system for processing.

Doctor : Doctor wants to be able to scan pictures of the ECG's and pass them to the system for processing..

Hospital : The hospital wants that all of their doctors and nurses can successfully scan ECG's on the app so that it can be further used for processing and detection of heart anomalies.

Patient: The patient wants that the nurse or doctor attending them is able to scan images of their ECG's for efficient heart anomalies detection.

Preconditions: The user must be logged in and authenticated

PostConditions: The user (nurse or the doctor) successfully scans the images and passes it to the system for further processing..

Main Success Scenario:

User Action	System Response
1 - User opens the mobile app	
2- The user is at the home screen (camera) after successful login.	
3. The user takes pictures of the printed ECG's based on the instructions given by the app.	
4.The user passes the images to the system for processing.	
	5. The system takes the input images and processes them.

Extensions:

1. At any time the system crashes : The user is then obliged to restart the app and scan the images again.

2.1.2.4 Use case name : Upload Images

Primary Actor : Nurse,Doctor

Scope : PulseViz App

Level : Primary

Stakeholders & Interests:

Nurse : Nurse wants to be able to upload pictures of already scanned ECG's and pass them to the system for processing.

Doctor : Doctor wants to be able to upload pictures of already scanned ECG's and pass them to the system for processing..

Hospital : The hospital wants that all of their doctors and nurses can successfully upload scanned ECG's on the app so that it can be further used for processing and detection of heart anomalies.

Patient: The patient wants that the nurse or doctor attending them is able to upload scanned images of their ECG's for efficient heart anomalies detection.

Preconditions : The user must be logged in and authenticated

PostConditions : The user (nurse or the doctor) successfully uploads the images and passes it to the system for further processing..

Main Success Scenario :

User Action	System Response
1 - User opens the mobile app	
2- The user is at the home screen (camera) after successful login.	
3. The user selects the 'upload' option	
4. The user selects images of the ECG from the gallery and hits upload.	
	5.The system first verifies if they meet the criterion .
	6. After successful verification of the images, they are then further processed.

Extensions :

1. At any time the system crashes : The user is then obliged to restart the app and upload the images again.
2. The uploaded images do not meet the criterion : The user is then told that the uploaded images weren't up to the criterion so they must upload some different pictures.

2.1.2.5 Use case name : View Results

Primary Actor : Nurse,Doctor

Scope : PulseViz App

Level : Primary

Stakeholders & Interests:

Nurse : Nurse wants to be able to view results of the ECG images so they can take further actions based on them.

Doctor : Doctor wants to be able to view results of the ECG images so they can take further actions based on them.

Hospital : The hospital wants that all of their doctors and nurses can successfully view ECG's results on the app so that it can be used to take further steps.

Patient: The patient wants that the nurse or doctor attending them is able to view results of their ECG for further procedure.

Preconditions : The user must be logged in and authenticated

PostConditions : The user (nurse or the doctor) successfully views the results of the images.

Main Success Scenario :

User Action	System Response
	1 - The system processes the images passed to it and detects any anomalies.
	2. The system passes the results based on the processing of the images back to the user.
	3.The system sends notification to relevant doctors in case of emergency(heart attacks).
4. The user views the results and takes further action	

Extensions :

1. At any time the system crashes : The user is then obliged to restart the app and scan the images again.

2.1.2.6 Use case name : View Report

Primary Actor : Nurse,Doctor

Scope : PulseViz App

Level : Primary

Stakeholders & Interests:

Nurse : Nurse wants to be able to view a detailed report of the patient's scanned ECG for better understanding of the whole situation.

Doctor : Doctor wants to be able to view a detailed report of the patient's scanned ECG for better understanding of the whole situation.

Hospital : The hospital wants that all of their doctors and nurses can successfully view the ECG reports on the app so that it can be used to have a better understanding of the patient's condition.

Patient: The patient wants that the nurse or doctor attending them is able to view reports of their ECG for efficient diagnosis and treatment.

Preconditions : The user must be logged in and authenticated

PostConditions : The user (nurse or the doctor) successfully views the report of the ECG.

Main Success Scenario :

User Action	System Response
	1. The system generates a report after carefully analyzing the scanned images.
	2 . The system then passes the report back to the user and displays it on their screens.
3. The user then views the reports generated by the app.	
4. The user then starts further medical procedures based on the report.	

Extensions :

1. At any time the system crashes : The user is then obliged to restart the app and upload the images again.

2.1.2.7 Use case name : Save Report

Primary Actor : Nurse,Doctor

Scope : PulseViz App

Level : Primary

Stakeholders and Interests:

Nurse : Nurse wants to be able to save the generated report in the systems database against that particular user so it can be used later for patient history.

Doctor : Doctor wants to be able to save the generated report in the systems database against that particular user so it can be used later for patient history.

Hospital : The hospital wants that all of their doctors and nurses can successfully save the report onto the database so that it can be viewed later as a patient's history.

Patient: The patient wants that the nurse or doctor attending them is able to save their report onto the database for later use and record.

Preconditions : The user must be logged in and authenticated

PostConditions : The report is successfully saved onto the database.

Main Success Scenario :

User Action	System Response
1 - User views the generated report.	
2- The user then clicks on “save report”..	
3. The user enters the required credentials for the report to be saved against and click “done”	
	4. The system saves the report into the database.
	5.The system then notifies the user about the success through a snackbar .

Extensions :

1. At any time the system crashes : The user is then obliged to restart the app and upload the images again.

2. The report is not saved due to uploading issues : The user is then notified about their current connectivity status and is asked to try to save again

2.1.2.8 Use case name : View Patient History

Primary Actor : Doctor

Scope : PulseViz App

Level : Primary

Stakeholders & Interests:

Doctor : Doctor wants to be able to view a patient's history for a better under- standing of the case.

Hospital : The hospital wants that all of their doctors can successfully view a patient's history for better understanding of the situation and better treatment of the patient.

Patient: The patient wants that the doctor attending them is able to view their patient history so that they can get a better diagnosis and treatment.

Preconditions : The user must be logged in and authenticated

PostConditions : The user (nurse or the doctor) successfully views the patient’s history on their mobile screen.

Main Success Scenario :

User Action	System Response
1 - User opens the mobile app	
2- The user then redirects to the page where it has the option to view patient history.	
3. The user clicks on view patient history.	
4. The user enters the required credentials and hit “search”.	
	5.The system grabs the patient’s history from the database.
	6. The system then displays the patient’s history on the mobile screen.
7. The user views the patient’s history and takes into consideration for further actions,	

Extensions :

1. At any time the system crashes : The user is then obliged to restart the app and try again.
2. Incorrect credentials or the record doesn’t exist : The user is then told to verify the credentials again and try , if they still see the same results then the patient’s

history does not exist in the system.

2.1.2.9 Use case name : Manage account

Primary Actor : Nurse,Doctor

Scope : PulseViz App

Level : Primary

Stakeholders and Interests:

Nurse : Nurse wants to be able to be able to view their profile, make changes or updates and even delete their account if they want to.

Doctor : Doctor wants to be able to be able to view their profile, make changes or updates and even delete their account if they want to.

Hospital : The hospital wants that all of their doctors and nurses can successfully view their accounts information , amend changes or delete the account so the system stays updated.

Preconditions : The user must be logged in and authenticated

PostConditions : The made changes are successfully reflected both in the app and in the database.

Main Success Scenario :

User Action	System Response
1 - User opens the mobile app	
2- The user then redirects to the profile screen.	
3. The user clicks on “edit”.	
4. The user then makes the wanted changes.	
5. The user then presses the “save” button.	
	6. The system makes the changes in the database.
	7. The system displays a success snackbar to the user.
	8. The system reflects the changes in the account information in real-time.
9. The user sees the reflected changes on the app .	

Extensions :

1. At any time the system crashes : The user is then obliged to restart the app and save the changes again.

2. Event Response Table

Following is the event-response table for PulseViz :

Event	Source	Response
User requests sign up	User(Doctor or Nurse)	The system registers the user on the system’s database.
Successful signup	System	The system redirects the user to the login screen.
User submits login form	User	The system verifies user information from the database.
Successful login	System	The system redirects the user to the home screen.
User scans/uploads images	User	System takes the images ,verifies the format and passes it for further processing.
Image passed for processing	System	System processes the image and passes it to the model for prediction and report generation.
Heart attack detected by the system	System	The system displays the results and reports to the user and notifies doctors.
Normal heart condition detected	System	The system displays the result and the report to the user
User saves the report against current patient	User	The system takes the report and saves it against the given patient credentials.
Doctor views patient history with patient credentials	User	The system verifies the credentials and displays the patient's history to the doctor.
User makes any changes to their own account	User	The system updates the information in the database with the given information.

2.2 Functional Requirements

The functional requirements for our project for each module are stated below:

2.2.1 Module 1: ECG Scanning

Following are the requirements for module 1:

1. The system shall allow users to take pictures of scans using their mobile camera within the app interface.

2. The system shall enable users to upload images of ECG scans from their mobile's gallery.

2.2.2 Module 2: Heart Attack Detection

Following are the requirements for module 2:

1. The system shall process the given ECG images and pass them to the Deep Learning model in the next phase for heart attack detection.

2. The system shall analyze the processed ECG images passed to it using Deep learning model and detect occurrences of heart attacks.

2.2.3 Module 3: Personalized Health Reports

Following are the requirements for module 3:

1. The system shall display the results of analysis, indicating whether a heart attack is detected,quickly after the image has been processed by the system.

2. The system shall generate a detailed report about the patient based on the prediction and detection of the given ECG and allow users to it.

2.2.4 Module 4: Generate Notifications and Results

Following are the requirements for module 4:

1. In case of a heart attack, the system shall automatically notify relevant doctors by sending them a report with the patient's location in the hospital and scan results.

2.2.5 Module 5: View Patient History

Following are the requirements for module 5:

1. The system shall let the user store each patient's report against their respective ID (SSN), allowing for future retrieval and review of the report by the doctor for better analysis , diagnosis and detection.

2. The system shall allow doctors(only) to view the medical history of the patient in the app using their respective ID(SSN).

2.2.6 Module 6: Cloud Integration and Data Management

Following are the requirements for module 6:

1. The system shall allow a nurse or doctor to register an account and log in. This process needs to be performed just once, unless the user logs out.

2. The system shall allow users to edit their account information, such as name, email, and password, as well as delete their account if they want to.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements of our project:

2.1.1 Reliability

The system should have an MTBF of at least 500 hours. This will be measured through the error and bug detection during the testing phase. The system failure is any unplanned downtime, app crash that results in the app being unusable for scanning ECG and detect- ing heart attacks , this will end up in delayed heart attack detection and treatment of the patient.The failure can be avoided by performing regular system health checks.The sys- tem will detect any abnormal system behavior or in case the user is unable to perform the required tasks, in case of failure , the system will restart within 10 minutes (maximum) to ensure smooth operation .

2.1.2 Usability

The system shall allow the users to scan the images and view results within just 3 interactions with the system . The system will also confirm from the user before sending images for processing , to ensure no accidental file uploads .In case of failure of uploading , the system will inform the user in 10 seconds and then the user can try again. The system will allow the doctor to view a patient's history (old reports) with just 3 interactions with the system .

2.1.3 Performance

95 % of the time , the system shall generate report(results) within 10 seconds of submis- sion under normal load circumstances. 99% of the time, the system shall notify doctors(in case of emergencies) within 3 seconds of the report generation given a stable internet connection . The system shall maintain an uptime for 99.5 % , measured over a period of 30 days , to ensure smooth operation . The system shall be able to retrieve previous records within 3 seconds of the request . The system shall be able to handle 100 concurrent users with minimum response delay.

2.1.4 Security

The system shall only allow access to the app to identified (registered) users. The system shall also enforce role-based restrictions and access to make sure only the desired people have the right to access certain private data . In case of 5 or more failed attempts to login within 5 minutes , the system shall lock the system , making it inaccessible to use on that particular device for a given time . The system shall also maintain logs, to keep track of the activity and where it’s originating from .

2.2 Domain Model

The domain model for Pulse-Viz is given below:

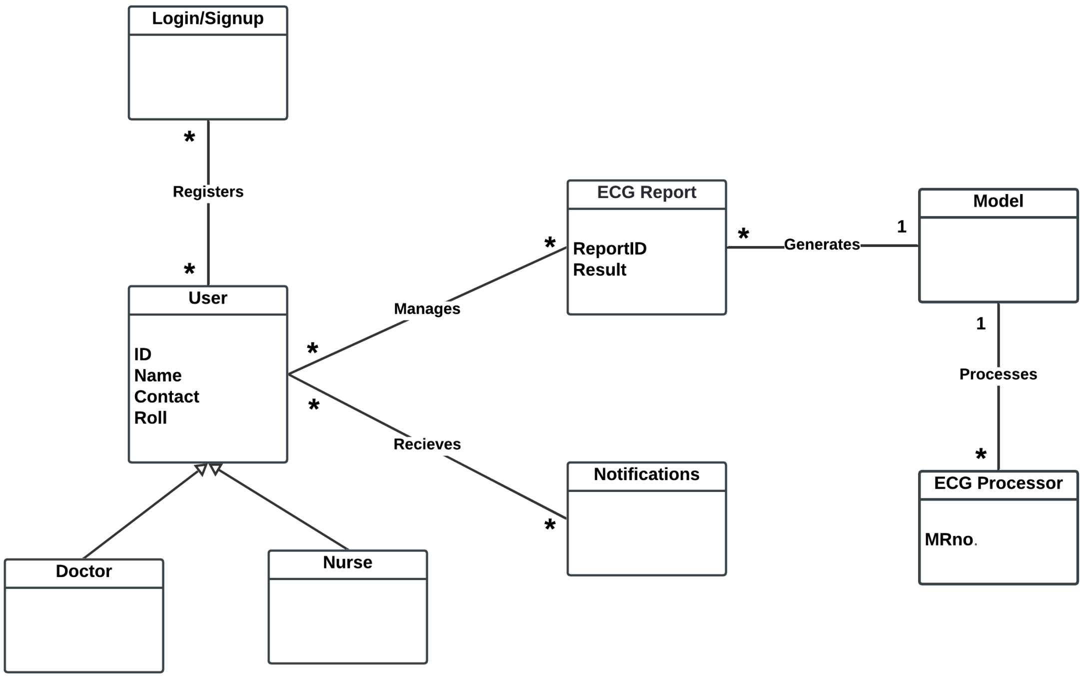


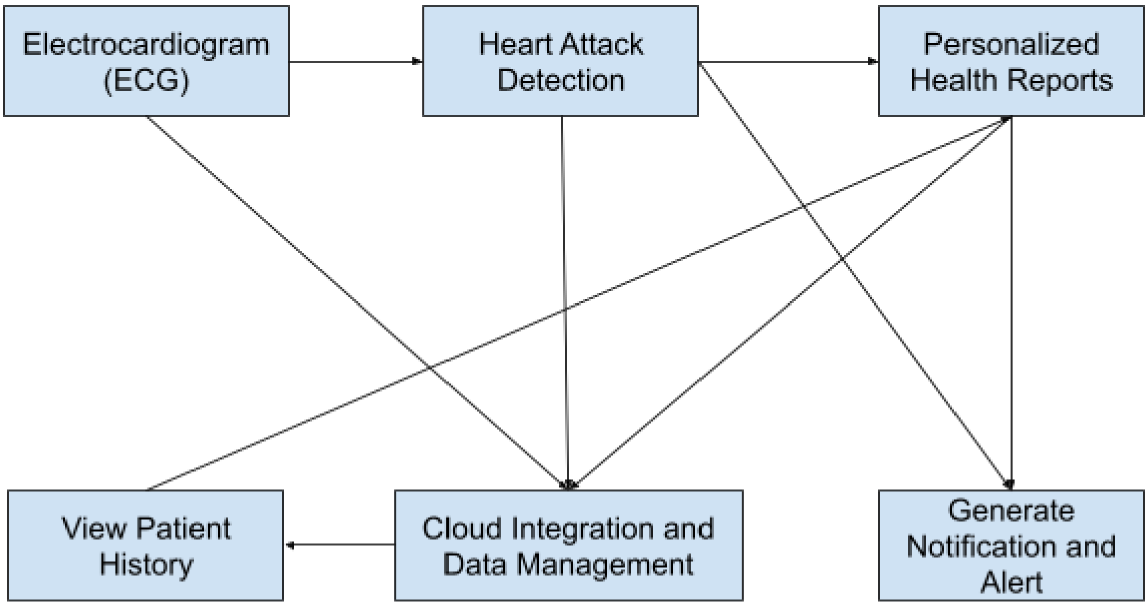
Figure 2.2: Domain Model

Chapter 3 System Overview

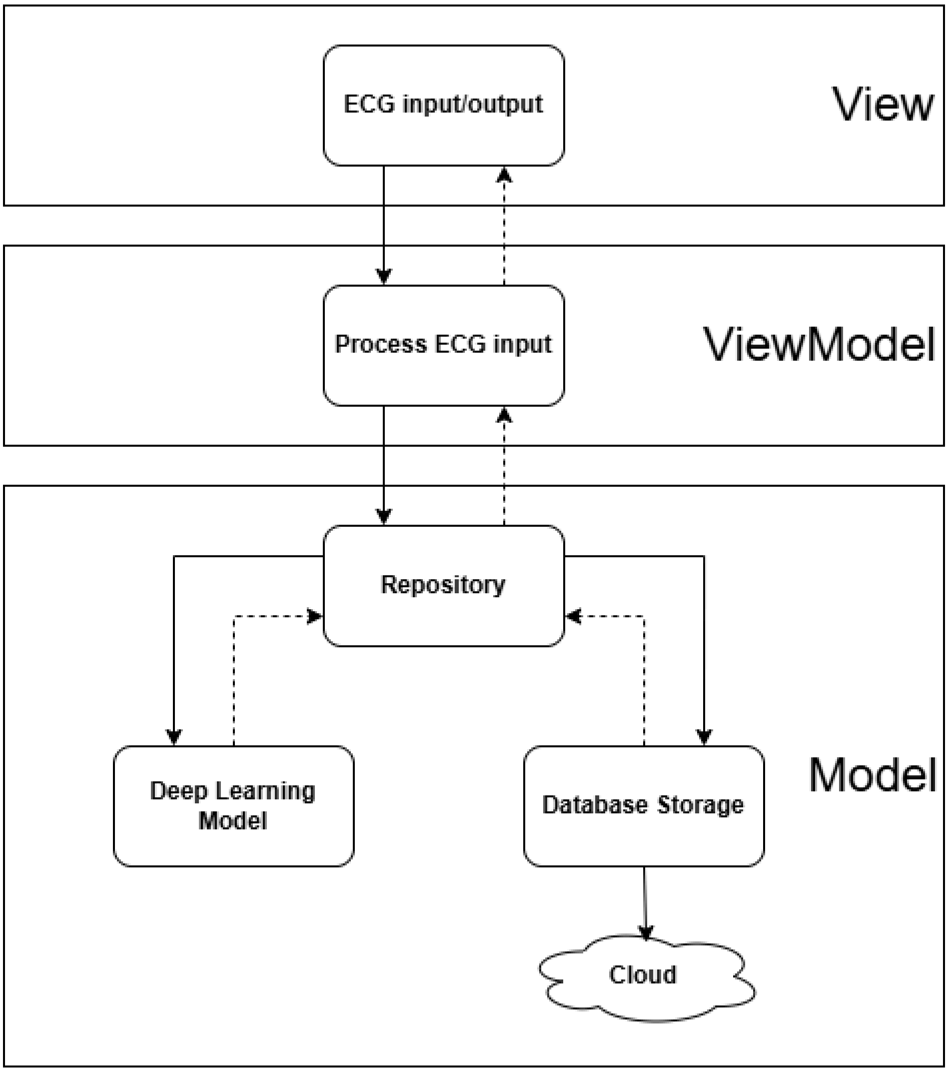
The PulseViz automates the analysis of ECG data to detect heart attacks in real-time. It processes input from scanned images or uploaded images, analyzes the data using advanced deep learning algorithms, and sends alerts when a heart attack is detected. The system is designed for hospitals in Pakistan, aiming to improve the speed and accuracy of heart attack diagnosis. It follows the MVVM architecture to maintain a clean separation between the user interface, data processing, and core logic.

3.1 Architectural Design [1]

3.1.1. Box Line Diagram



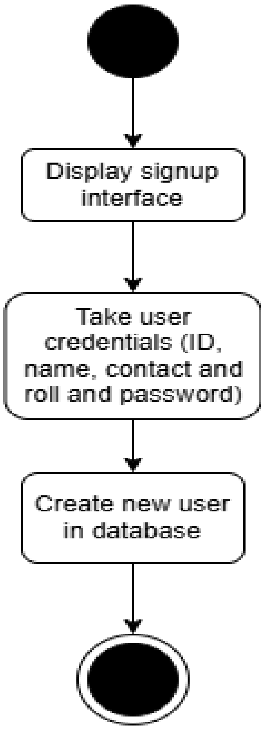
3.1.2. Architecture Diagram- MVVM Architecture



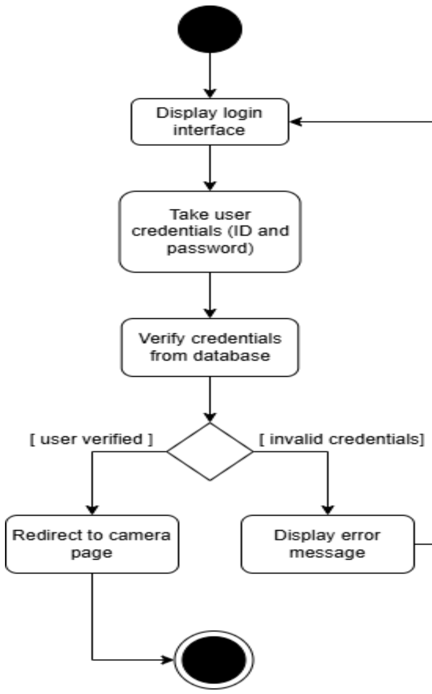
3.2 Design Models

3.2.1. Activity Diagram

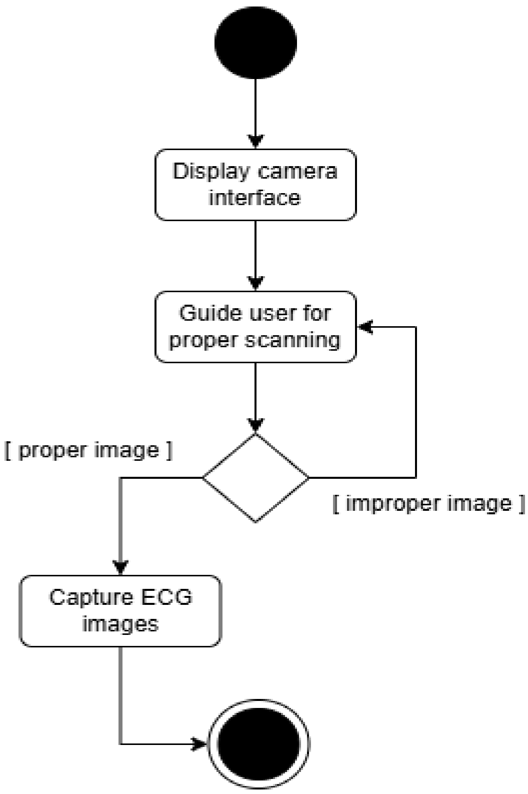
Use Case 1: Register new user



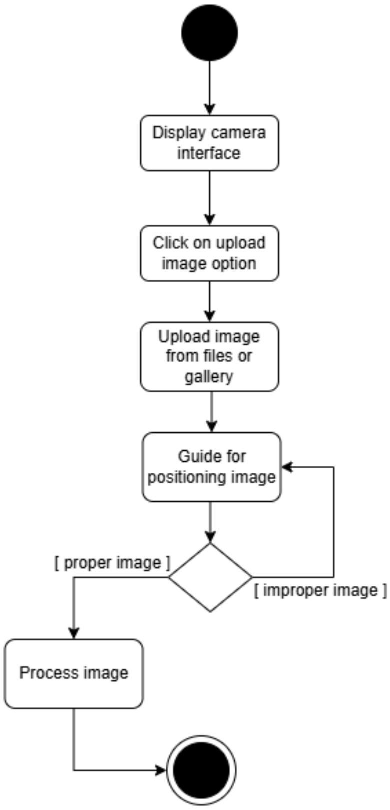
Use Case 2: Login existing user



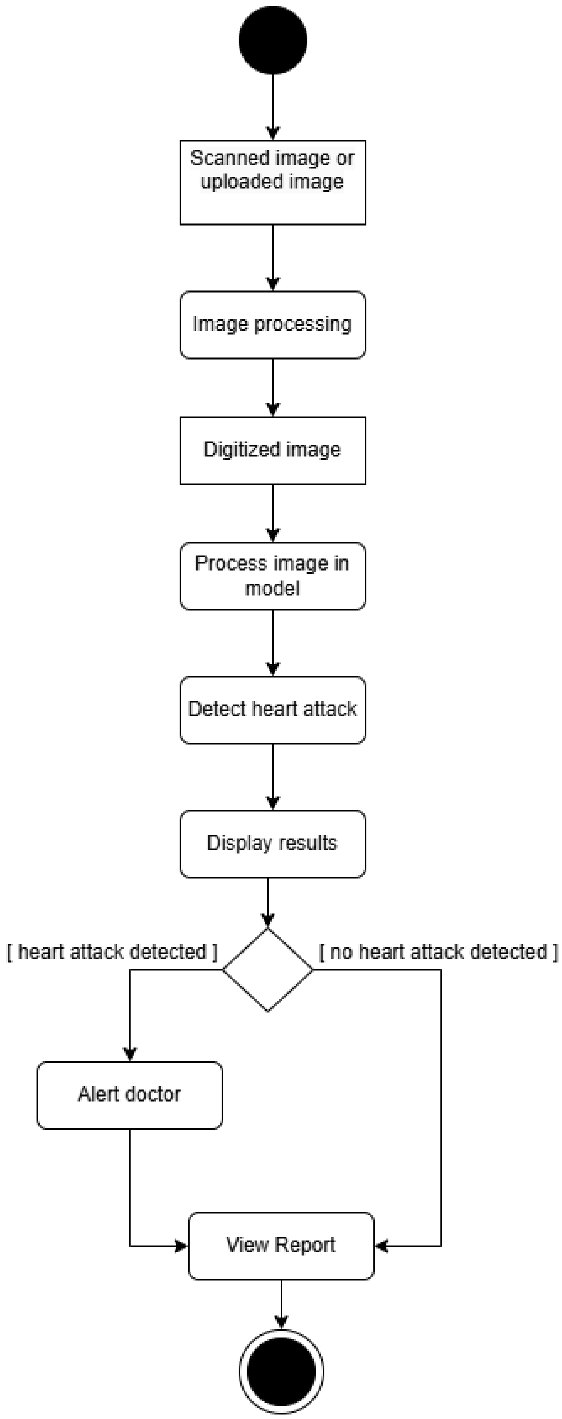
Use Case 3: Scan Images



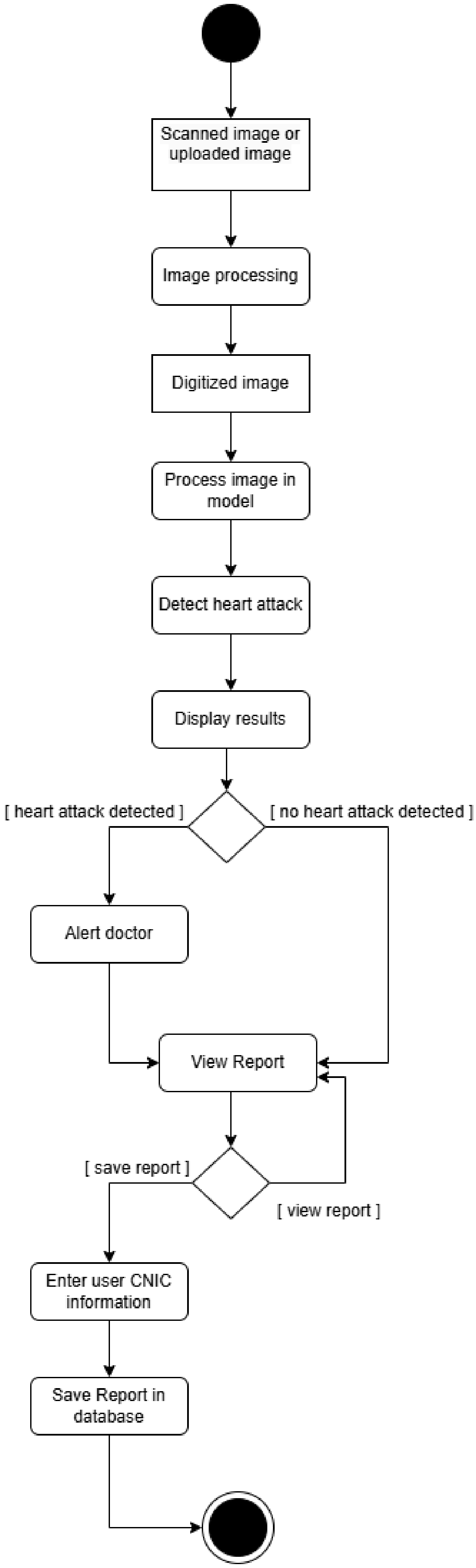
Use Case 4: Upload Images



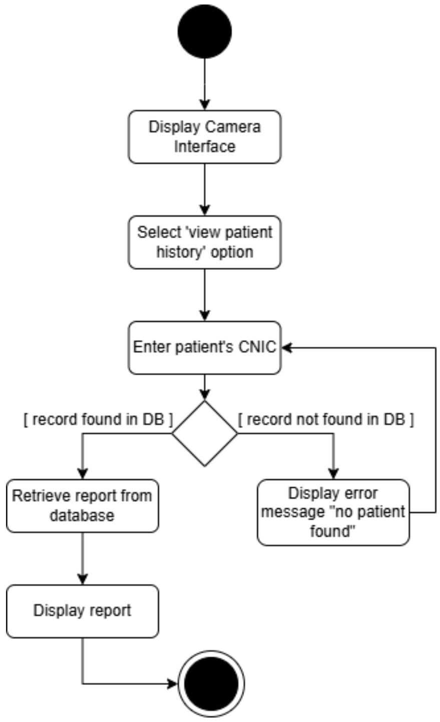
Use Case 5,6: View Results and Report



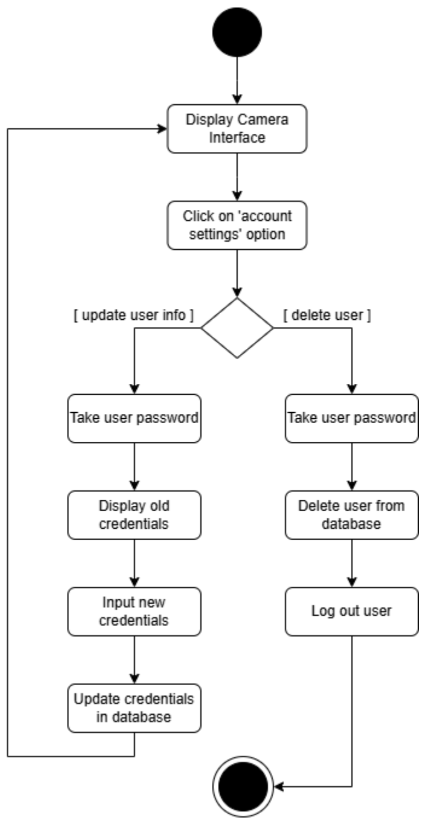
Use Case 7: Save Report



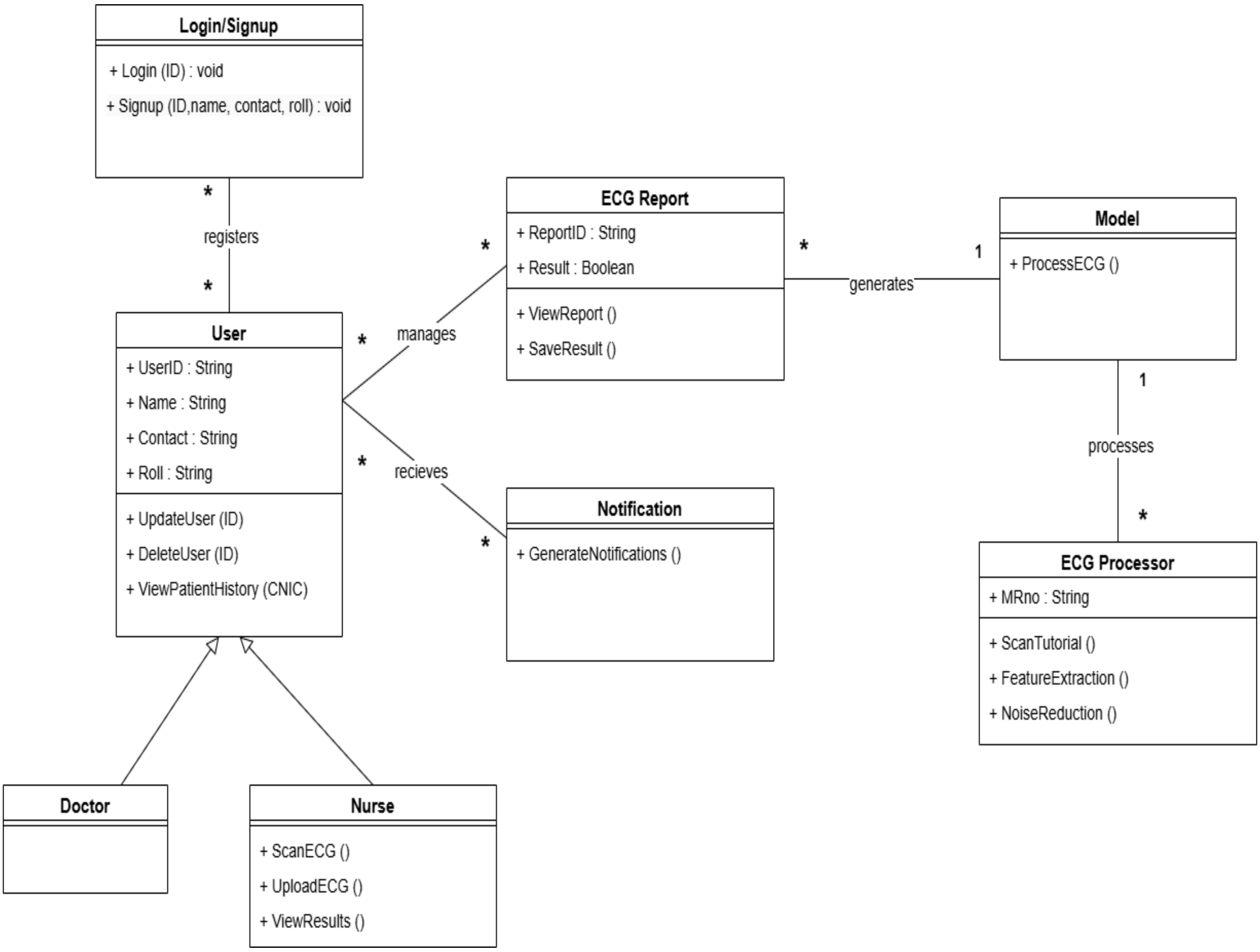
Use Case 8: View Patient History



Use Case 9: Manage Account

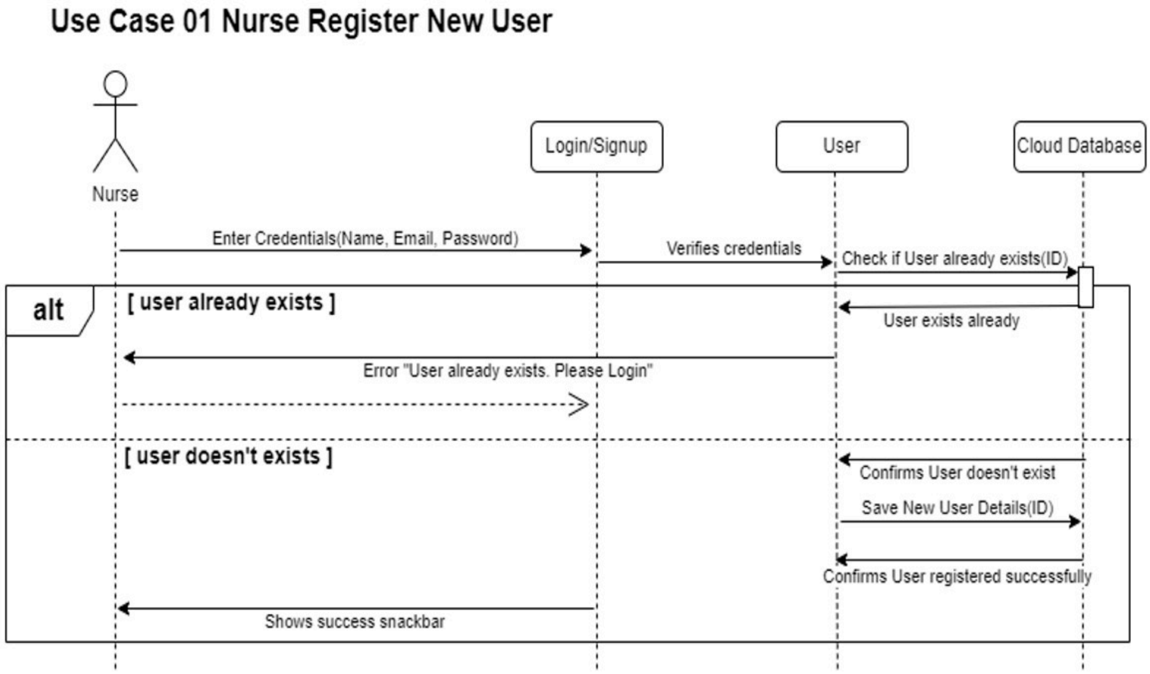


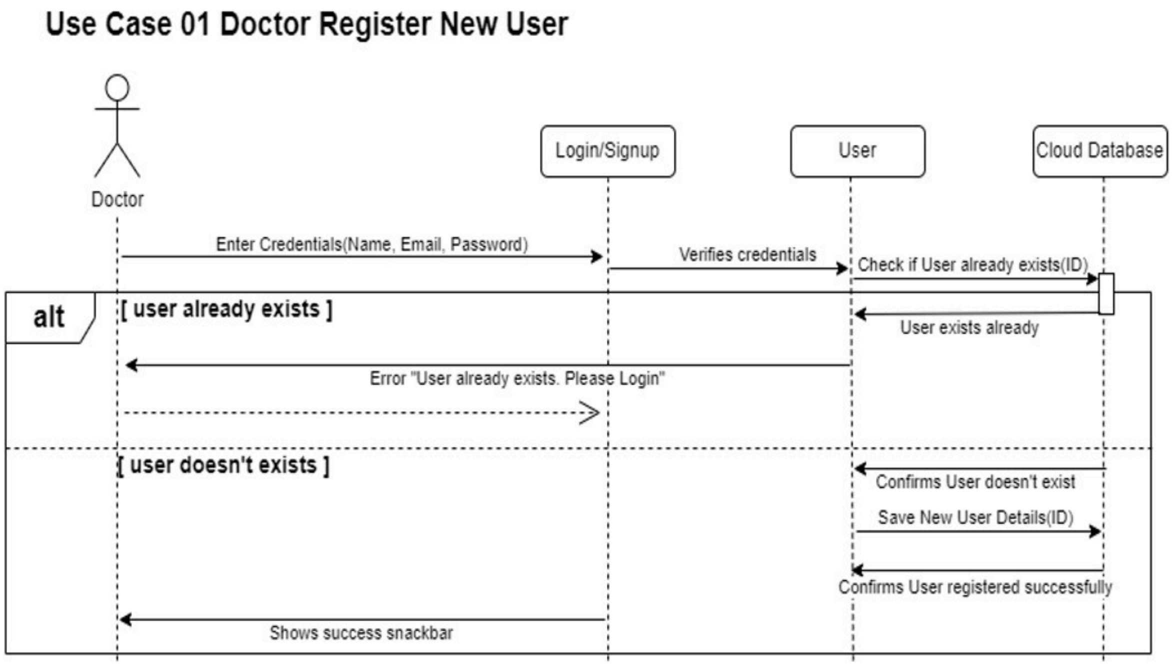
3.2.2. Class Diagram [3]



3.2.3. Class Level Sequence Diagram [2]

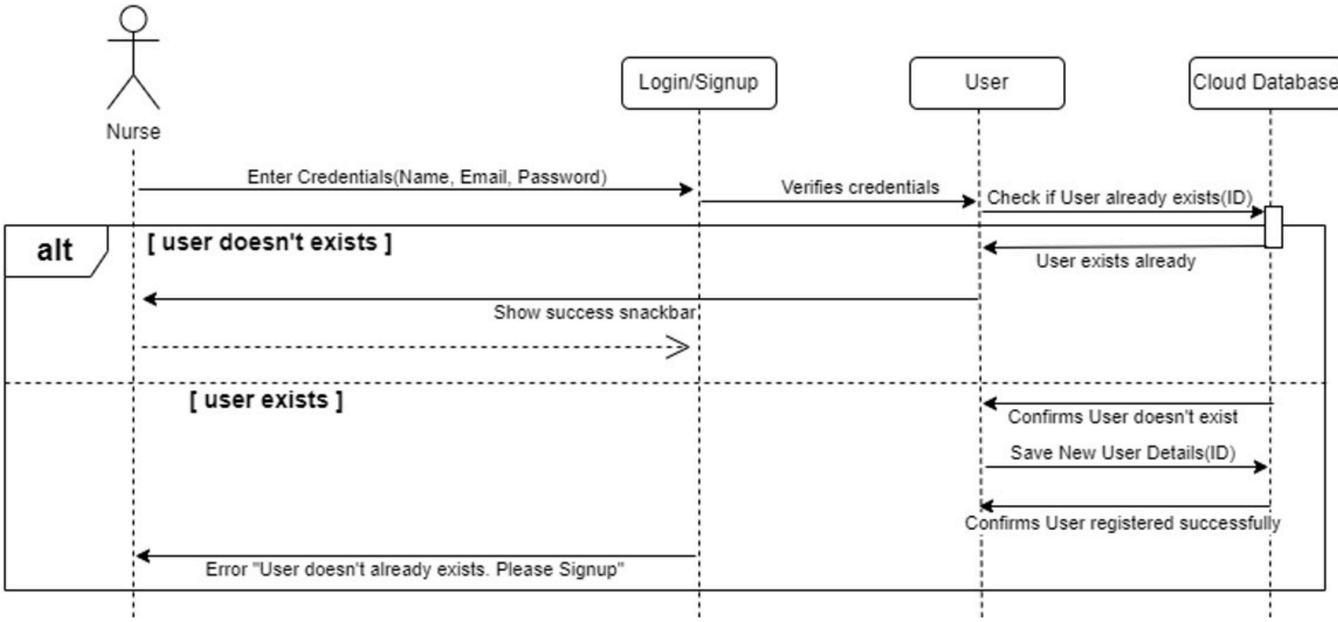
Use Case 1: Login existing user



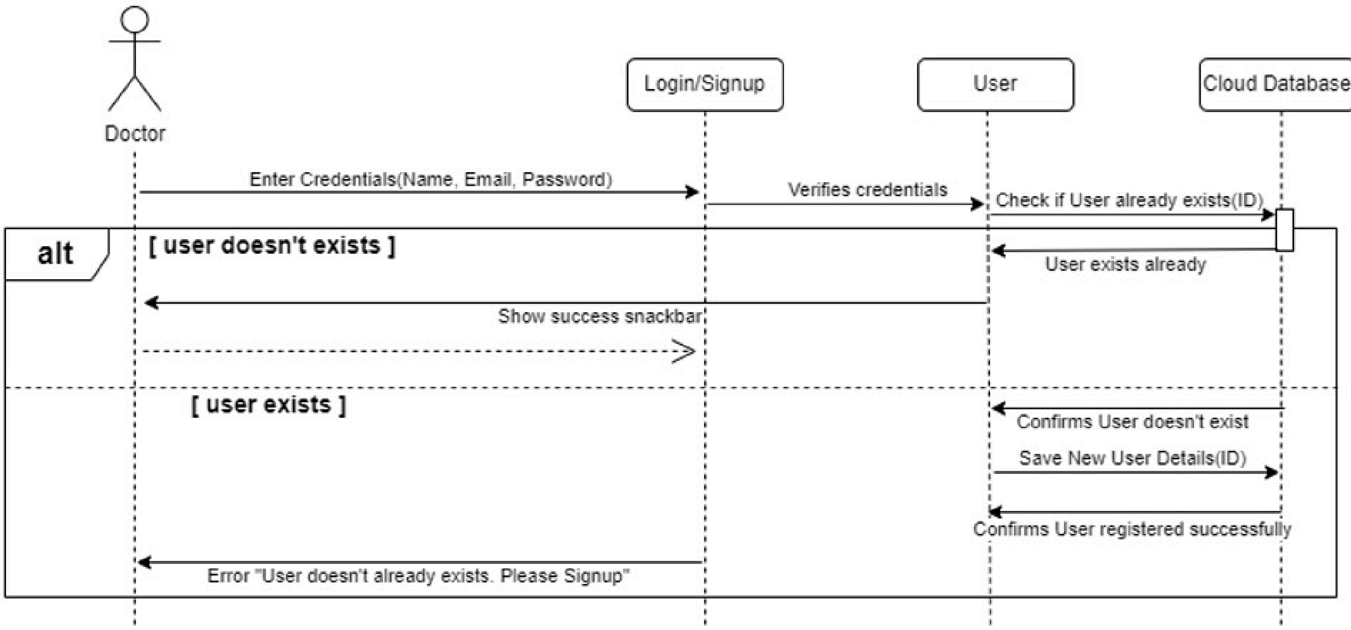


Use Case 2: Login existing user

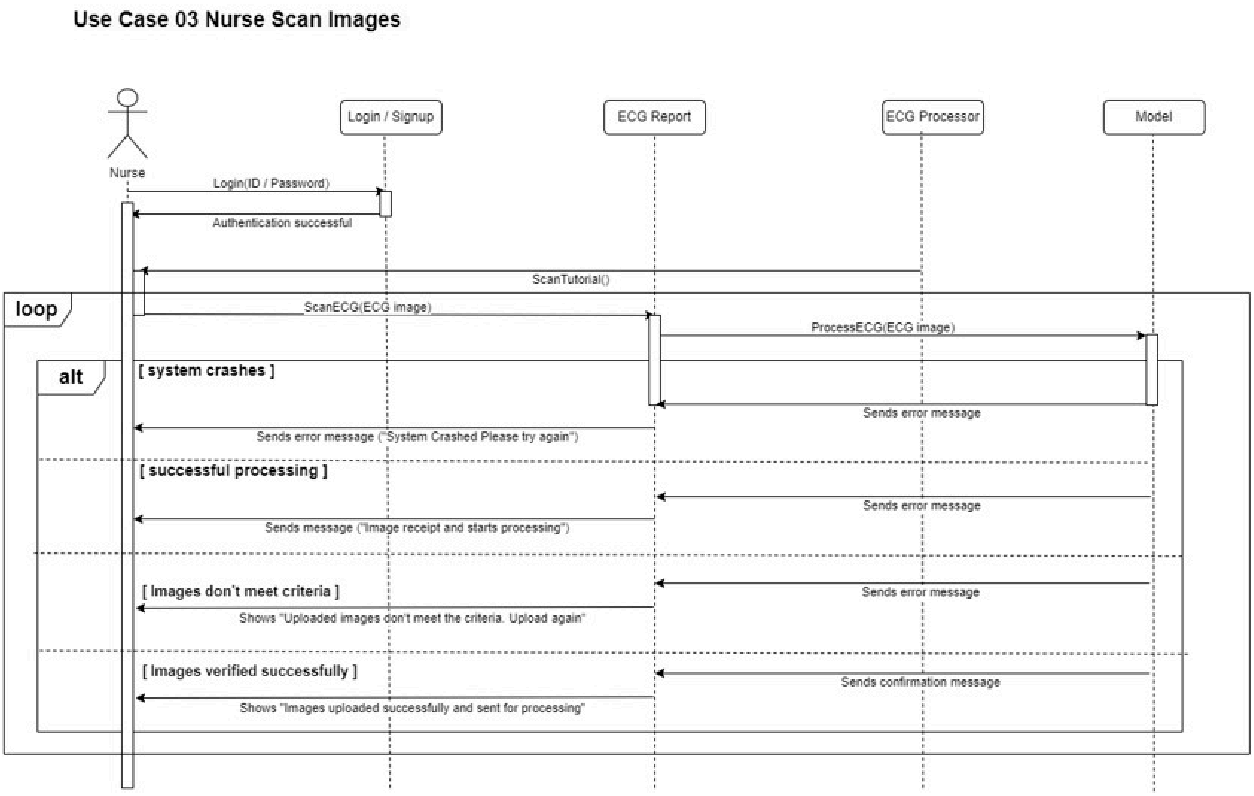
Use Case 02 Nurse Logins User



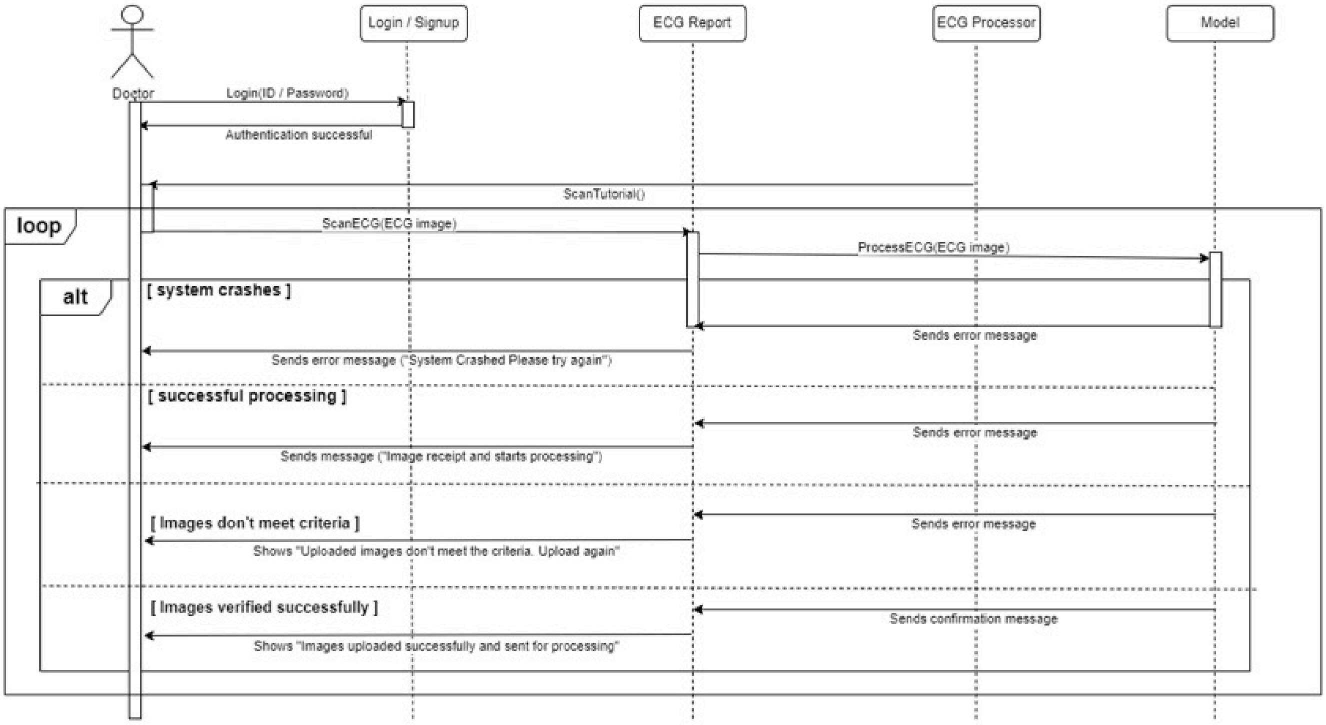
Use Case 02 Doctor Logins User



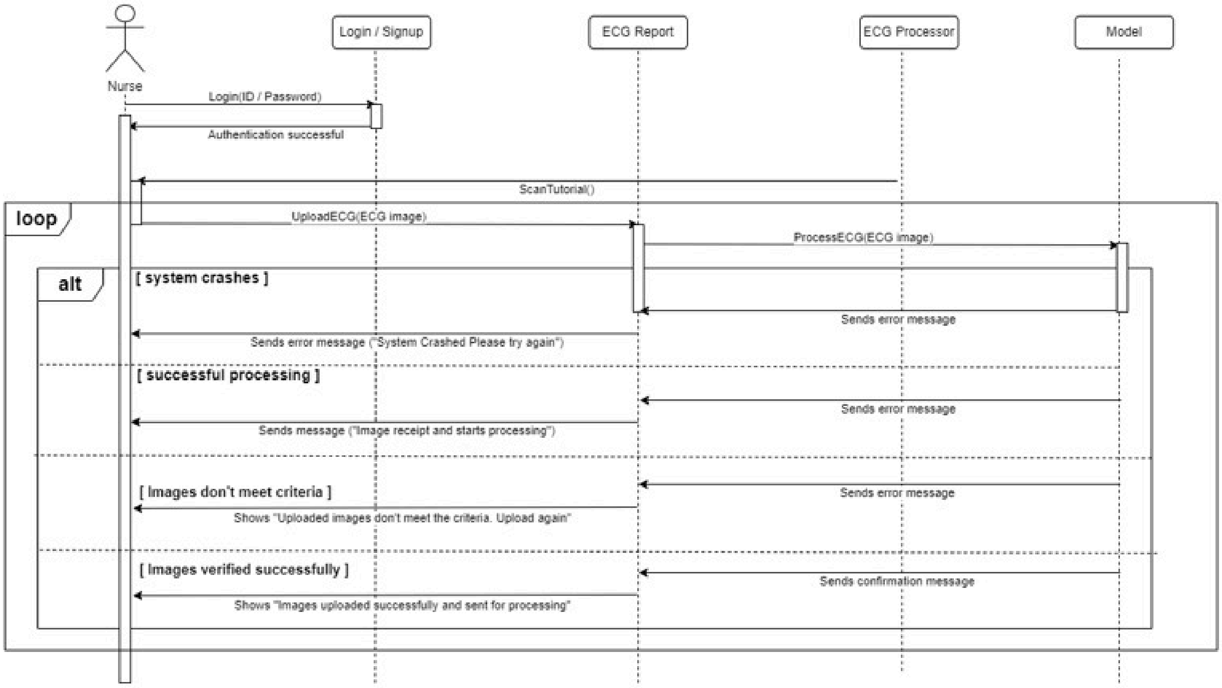
Use Case 3: Scan Images



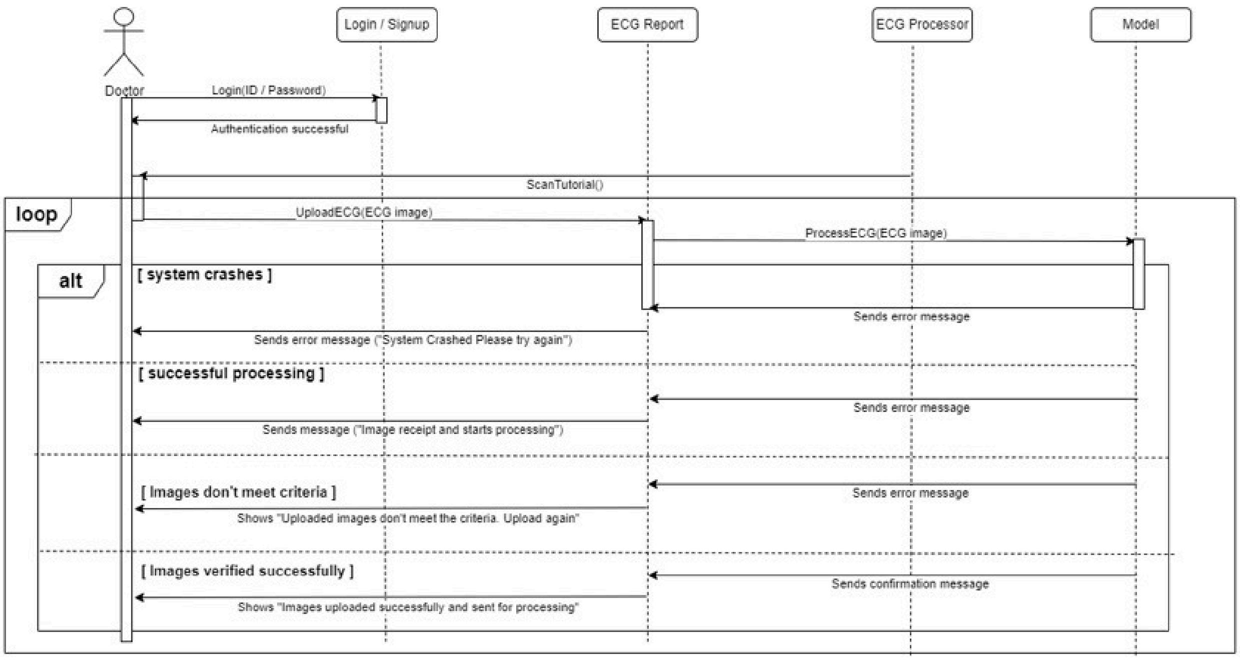
Use Case 03 Doctor Scan Images



Use Case 4: Upload Images



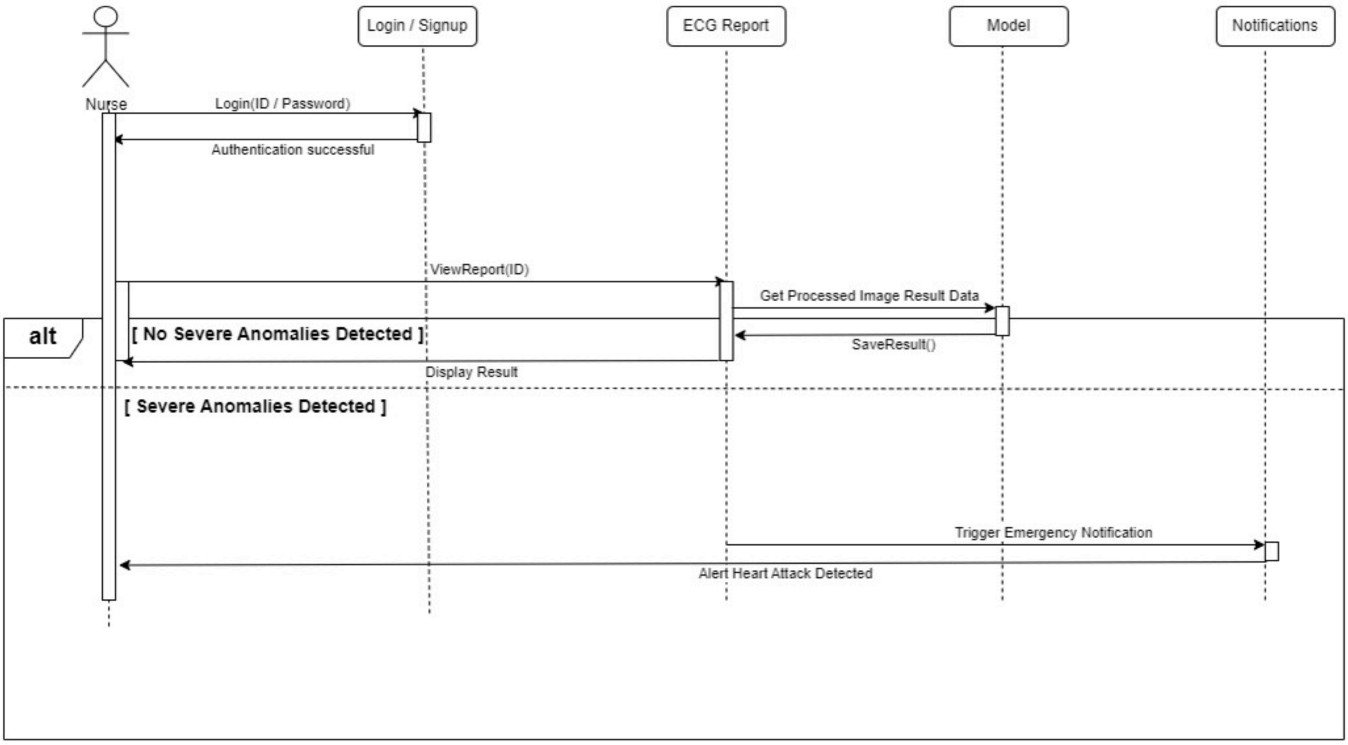
Nurse
Login / Signup
ECG Report
ECG Processor
Model
Login(ID / Password)
Authentication successful
ScanTutorial()
loop
UploadECG(ECG image)
ProcessECG(ECG image)
alt
[system crashes]
Sends error message
Sends error message ("System Crashed Please try again")
[successful processing]
Sends error message
Sends message ("Image receipt and starts processing")
Sends error message
[Images don't meet criteria]
Shows "Uploaded images don't meet the criteria. Upload again"
[Images verified successfully]
Sends confirmation message
Shows "Images uploaded successfully and sent for processing"



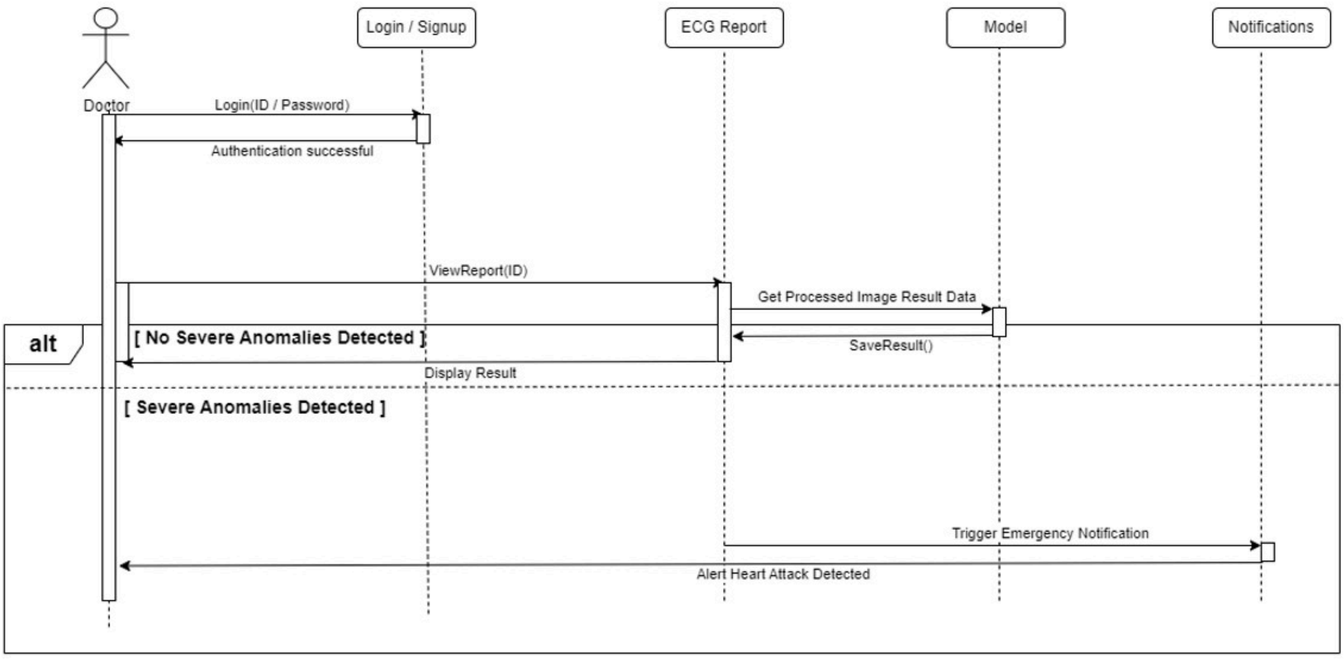
Doctor
Login / Signup
ECG Report
ECG Processor
Model
Login(ID / Password)
Authentication successful
ScanTutorial()
loop
UploadECG(ECG image)
ProcessECG(ECG image)
alt
[system crashes]
Sends error message
Sends error message ("System Crashed Please try again")
[successful processing]
Sends error message
Sends message ("Image receipt and starts processing")
Sends error message
[Images don't meet criteria]
Shows "Uploaded images don't meet the criteria. Upload again"
[Images verified successfully]
Sends confirmation message
Shows "Images uploaded successfully and sent for processing"

Use Case 5: View Results

Use Case 05 Nurse View Results

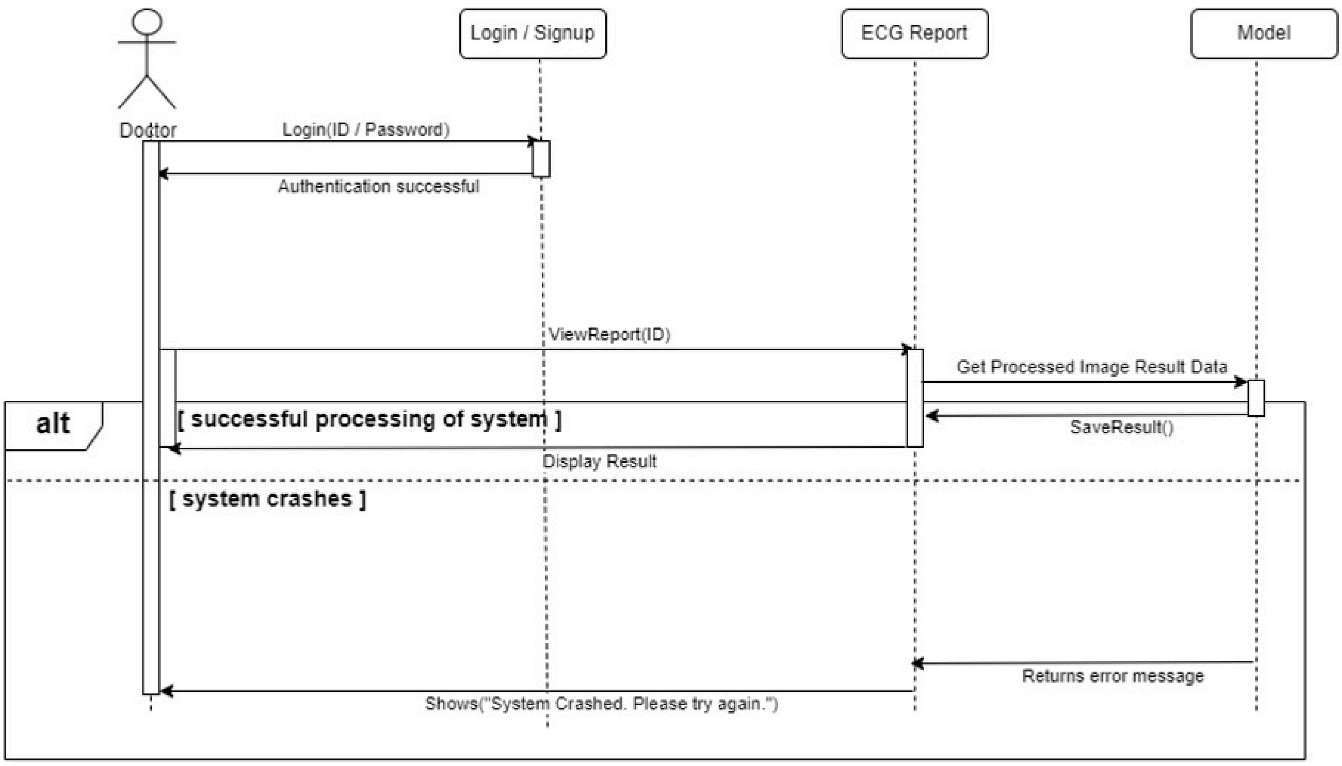


Use Case 05 Doctor View Result

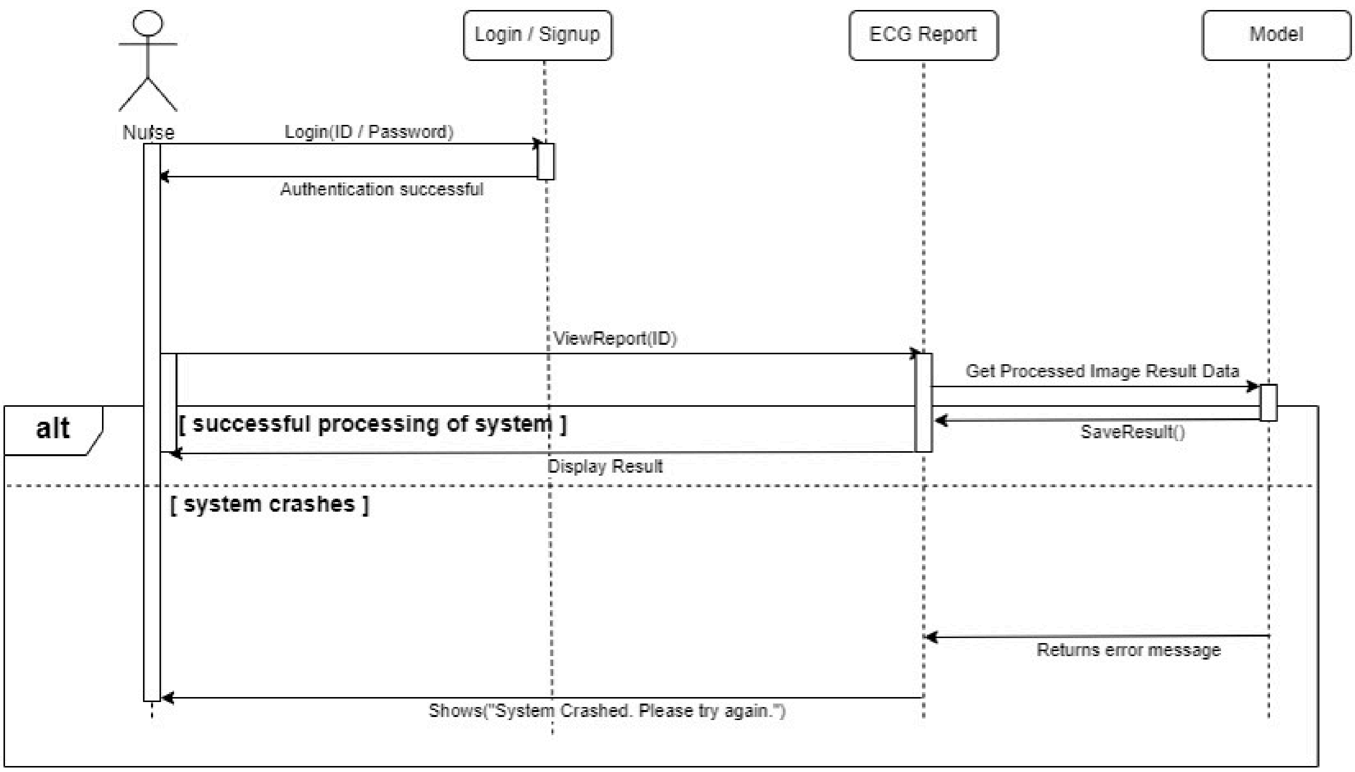


Use Case 6: View Report

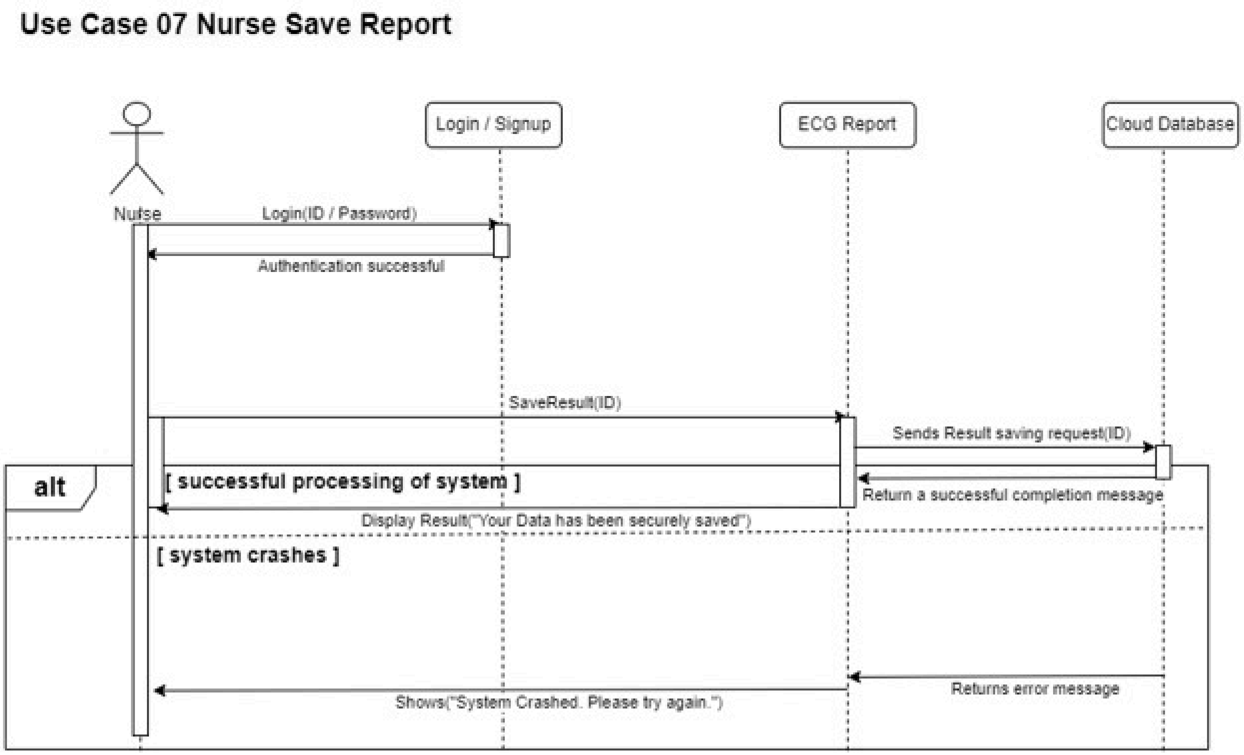
Use Case 06 Doctor View Report

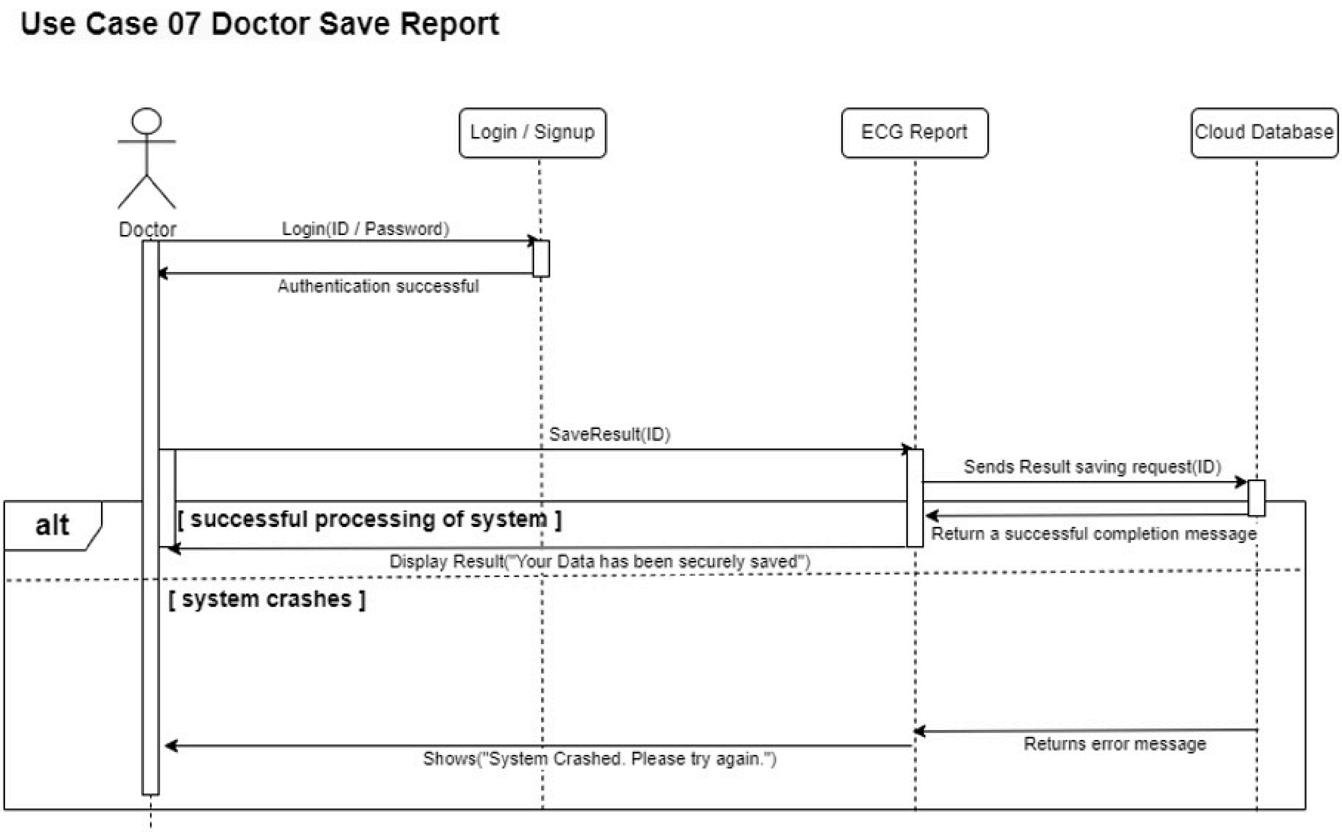


Use Case 06 Nurse View Report

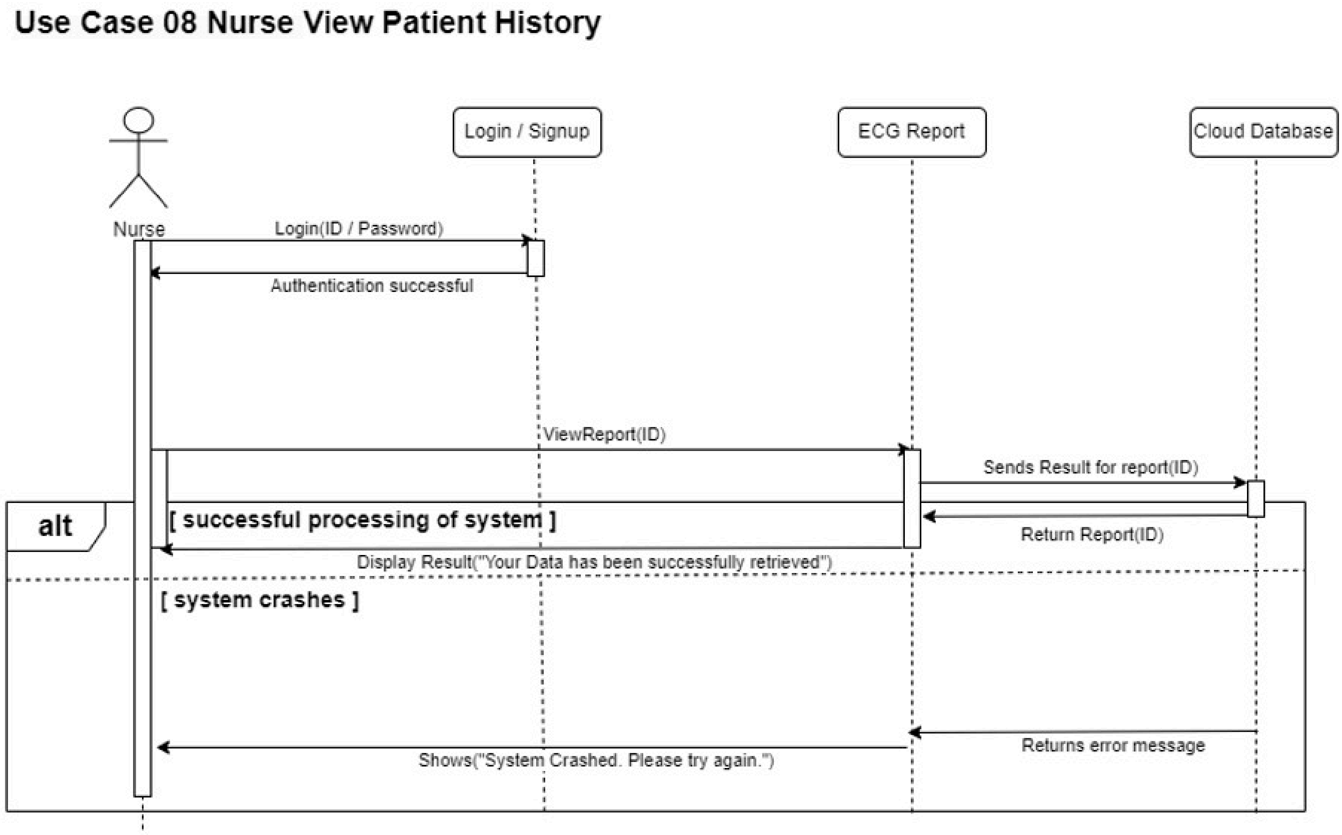


Use Case 7: Save Report

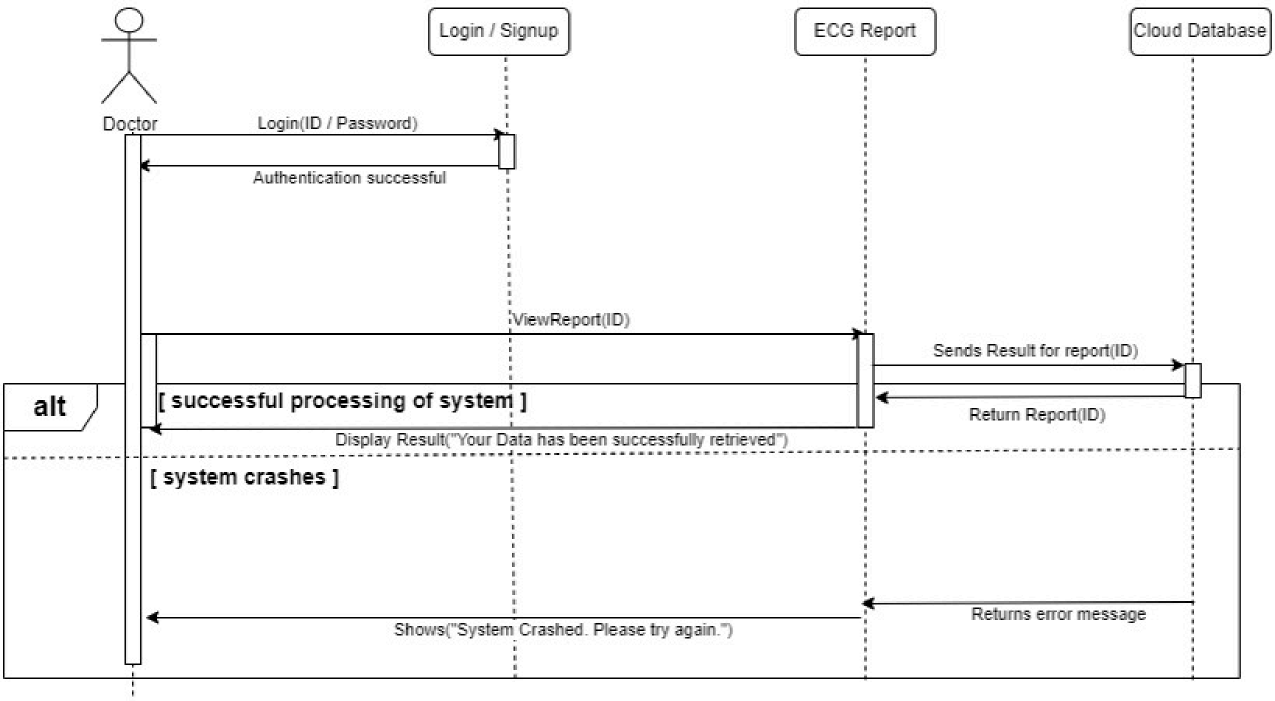




Use Case 8: View Patient History

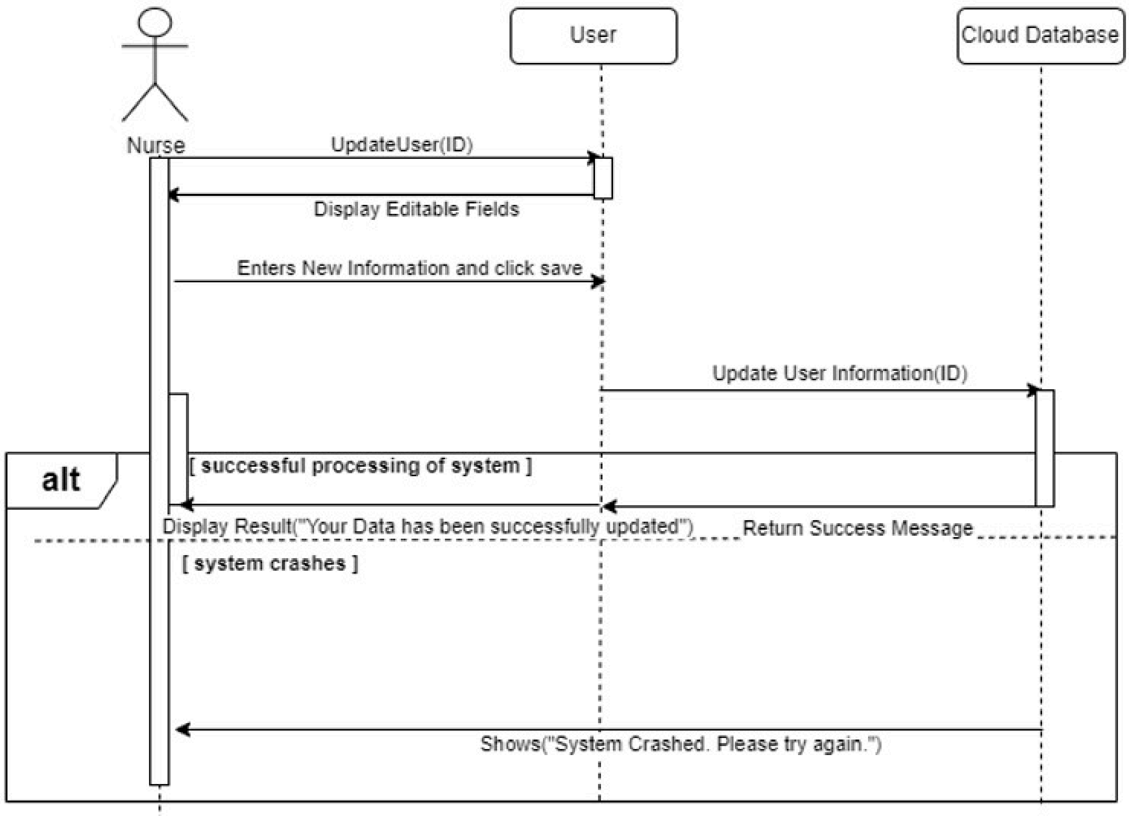


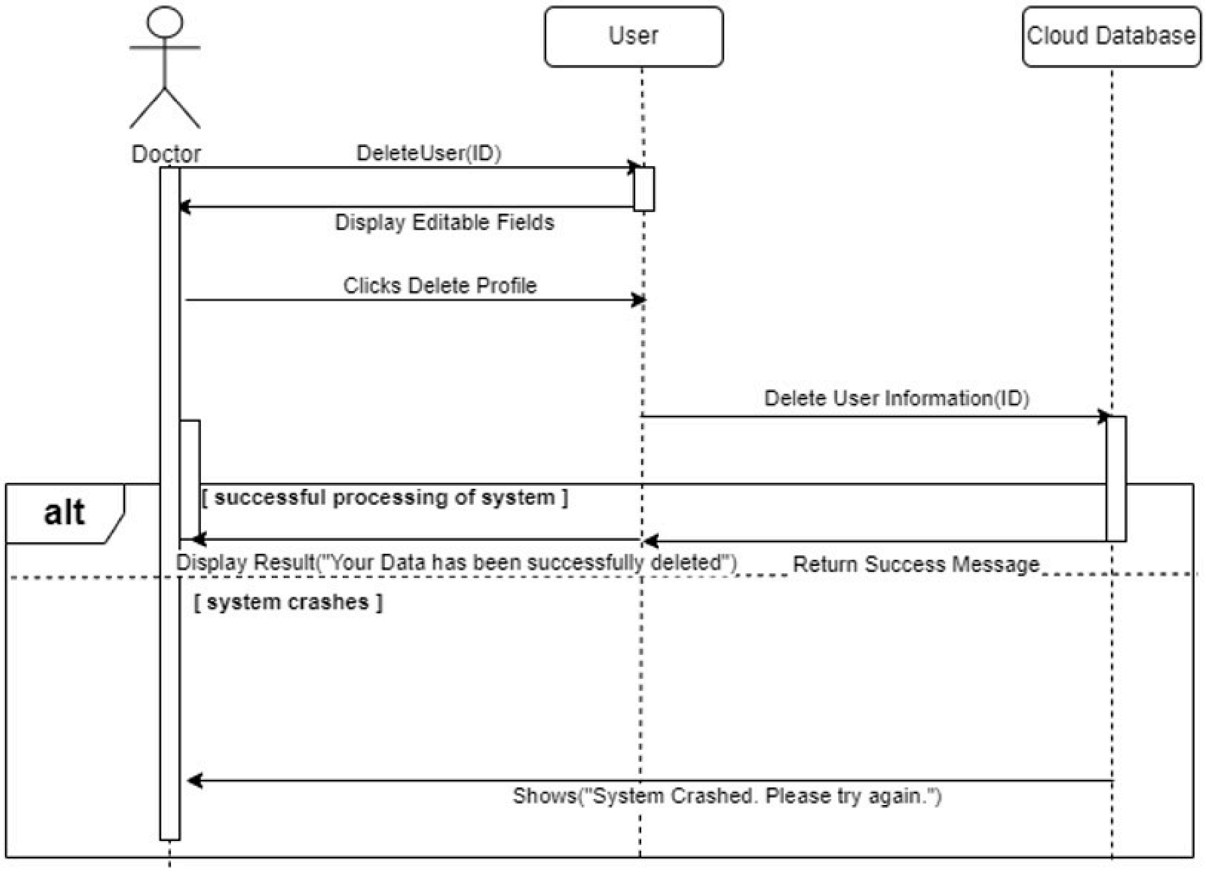
Use Case 08 Doctor View Patient History



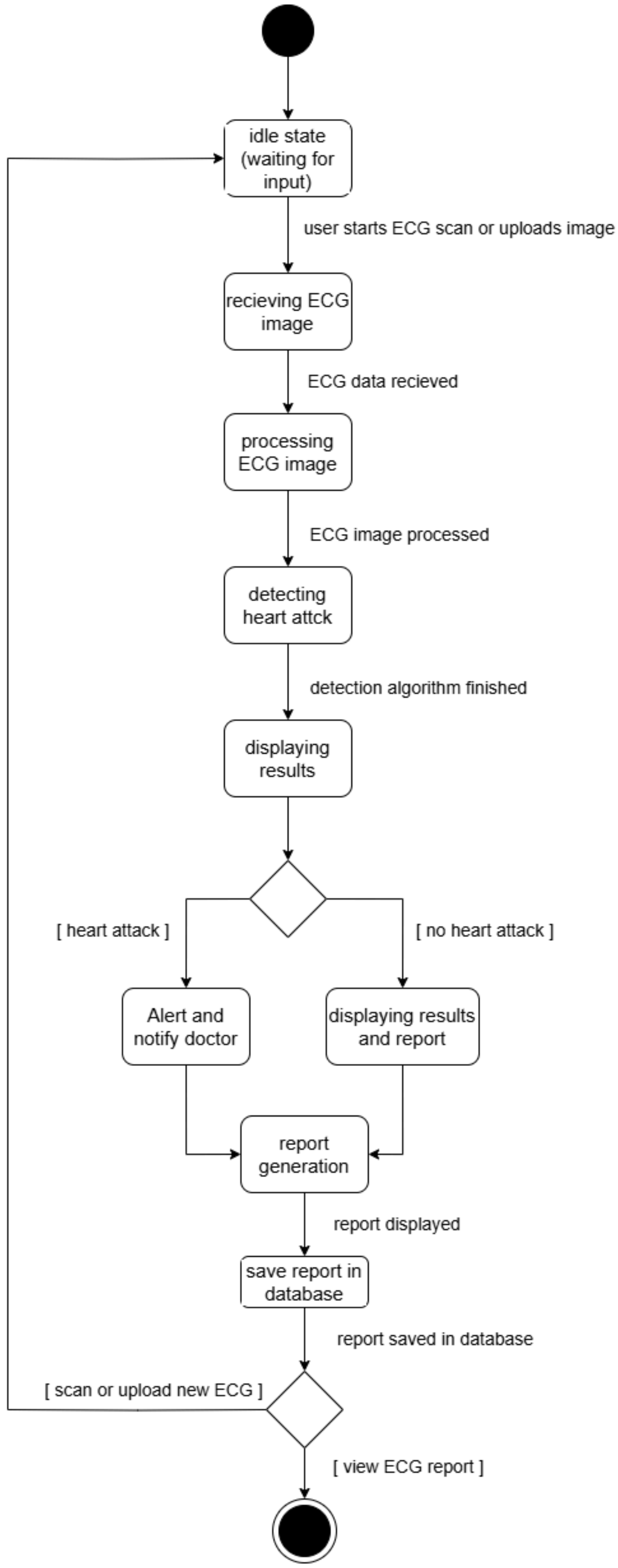
Use Case 9: Manage account

Use Case 09 Manage Account



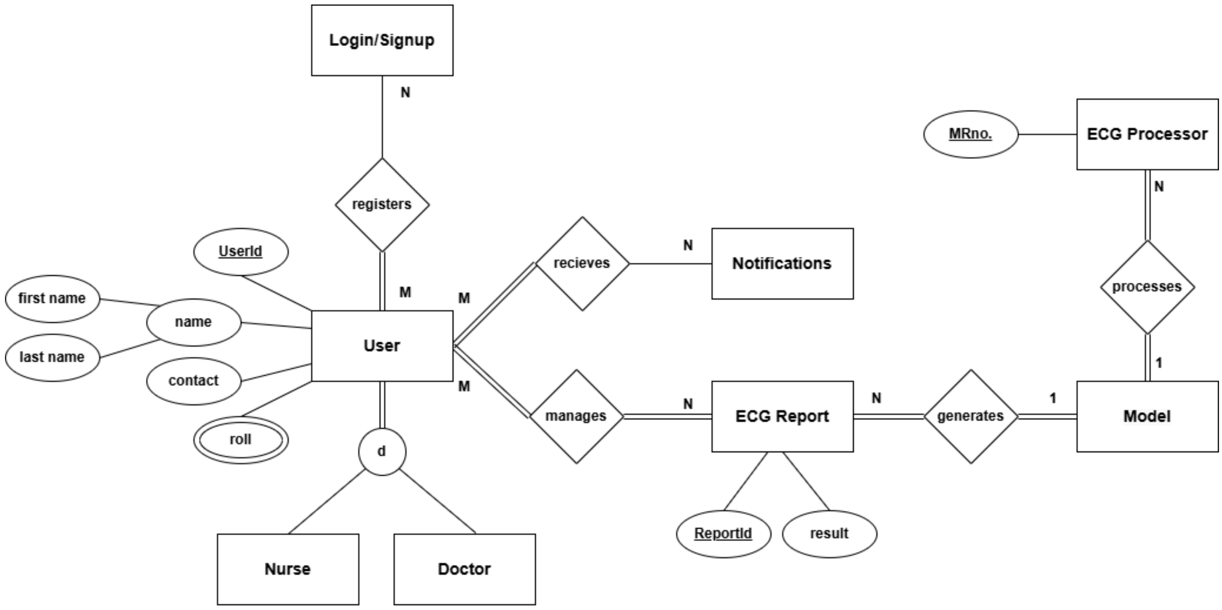

User
Cloud Database
Doctor
DeleteUser(ID)
Display Editable Fields
Clicks Delete Profile
Delete User Information(ID)
alt
[successful processing of system]
Display Result("Your Data has been successfully deleted")
Return Success Message
[system crashes]
Shows("System Crashed. Please try again.")

3.2.3. State Transition Diagram

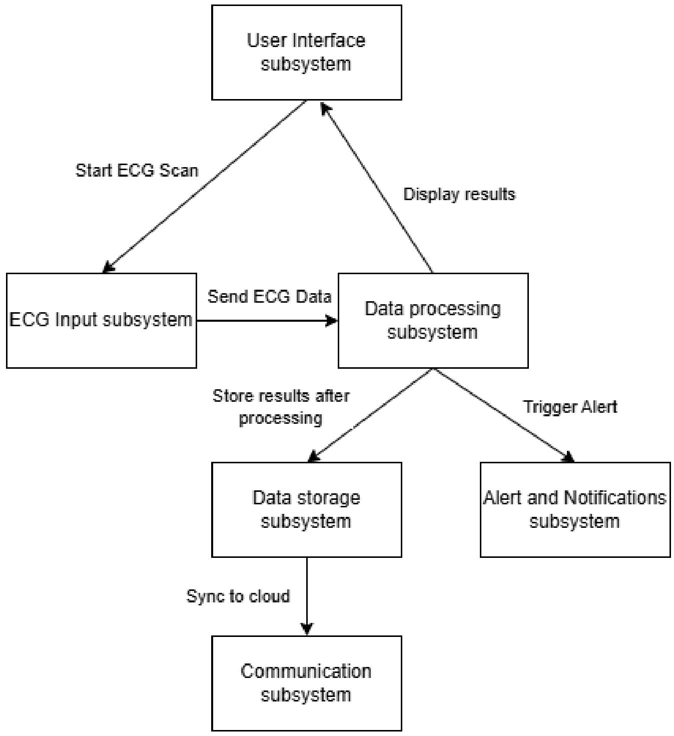


3.3 Data Design

This is the Entity Relationship diagram of our project, which depicts the relation between the classes in our application. This depicts the attributes of each class and how the classes are linked to each other. The database tables will include tables for users-doctor and nurse, ECG reports and patient's credentials.



Below listed are the subsystem connections of our project and how they interact with each other.
The diagram below shows what each subsystem sends or receives from the other subsystem, depicting the flow of information and processes.


User Interface subsystem
Start ECG Scan
Display results
ECG Input subsystem
Send ECG Data
Data processing subsystem
Store results after processing
Trigger Alert
Data storage subsystem
Alert and Notifications subsystem
Sync to cloud
Communication subsystem

Bibliography

[1]“What Is Architecture Diagramming? - Software & System Architecture Diagramming Explained - AWS.” *Amazon Web Services, Inc.*, aws.amazon.com/what-is/architecture-diagramming/.

[2] “[Wikipedia Contributors. “Sequence Diagram.” *Wikipedia*, Wikimedia Foundation, 12 Dec. 2019, en.wikipedia.org/wiki/Sequence_diagram.]”

[3] “[Rani, Bhumika. “Unified Modeling Language (UML) | Class Diagrams - GeeksforGeeks.” *GeeksforGeeks*, 30 Aug. 2018, www.geeksforgeeks.org/unified-modeling-language-uml-class-diagrams/.
]”